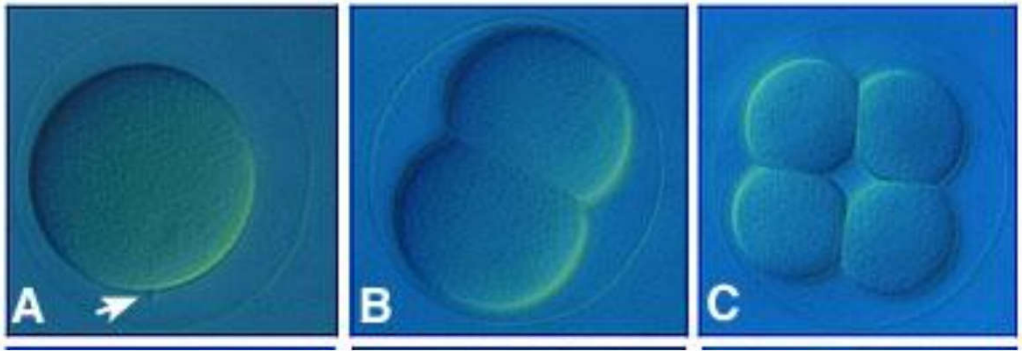
Chapter II: DNA replication
Mitosis and meiosis

Cell division and DNA replication

- Cells divide

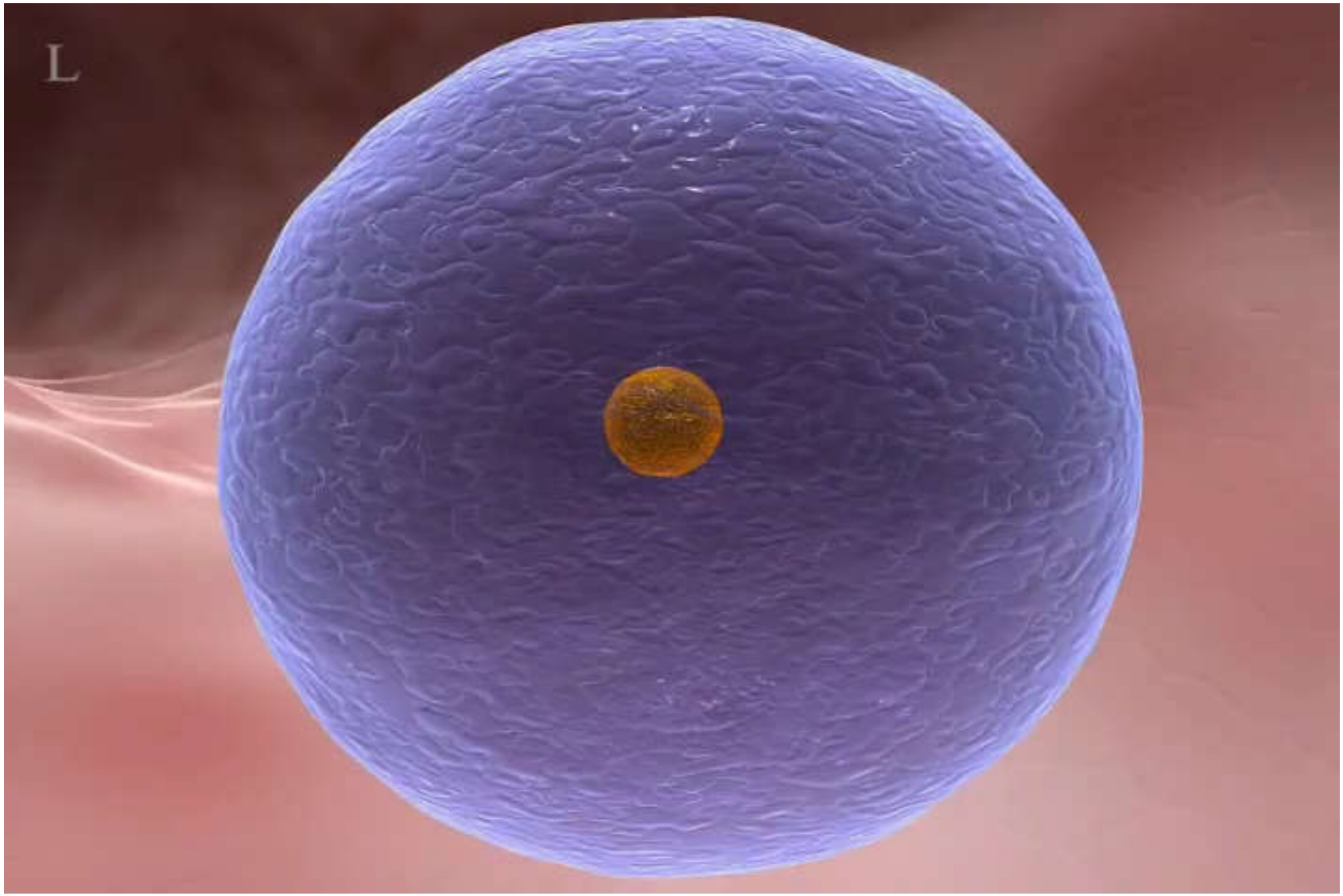
  ⇨Growth, Repair, Replacement



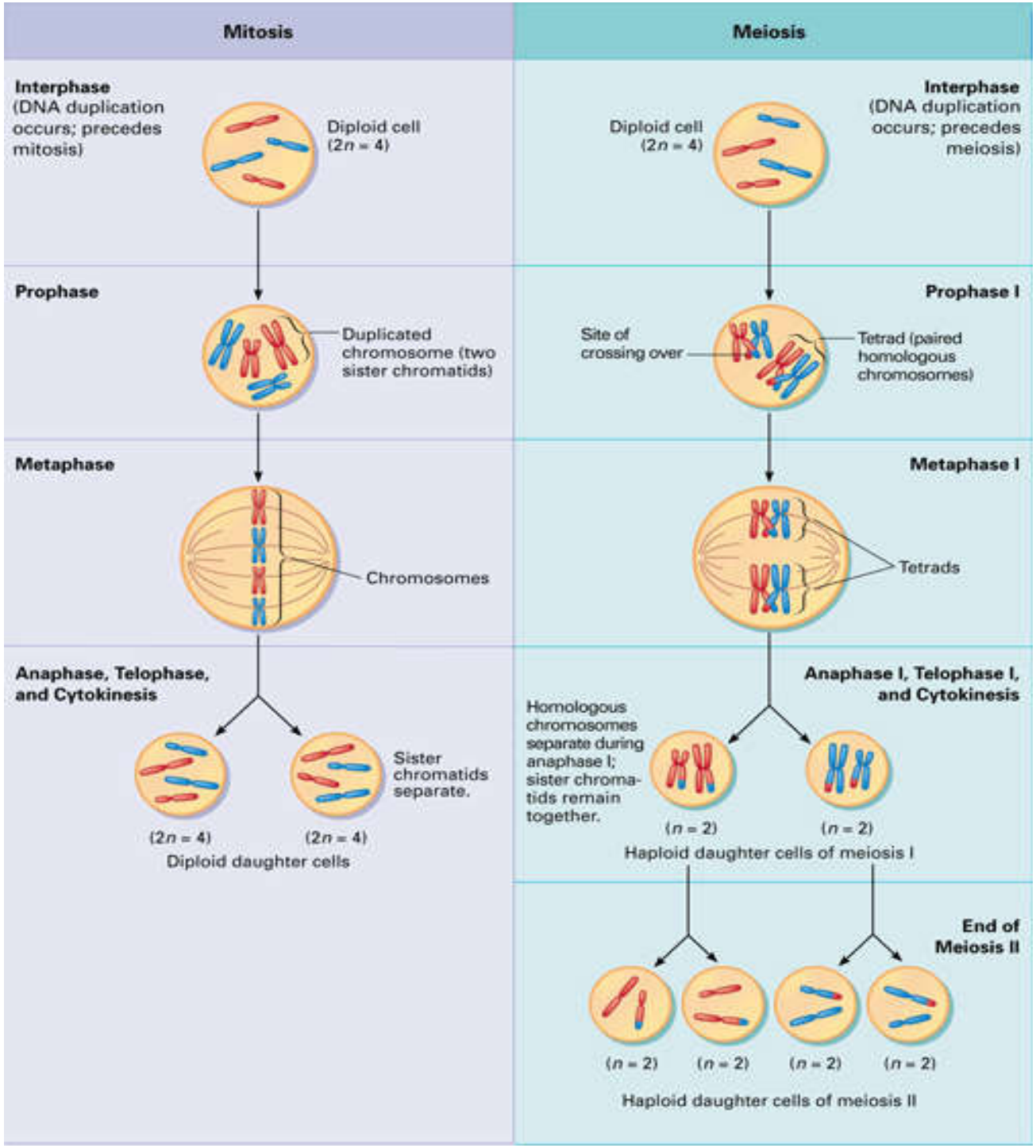
- Before cells divide they have to double cell structures, organelles and their genetic information



Embryonic development



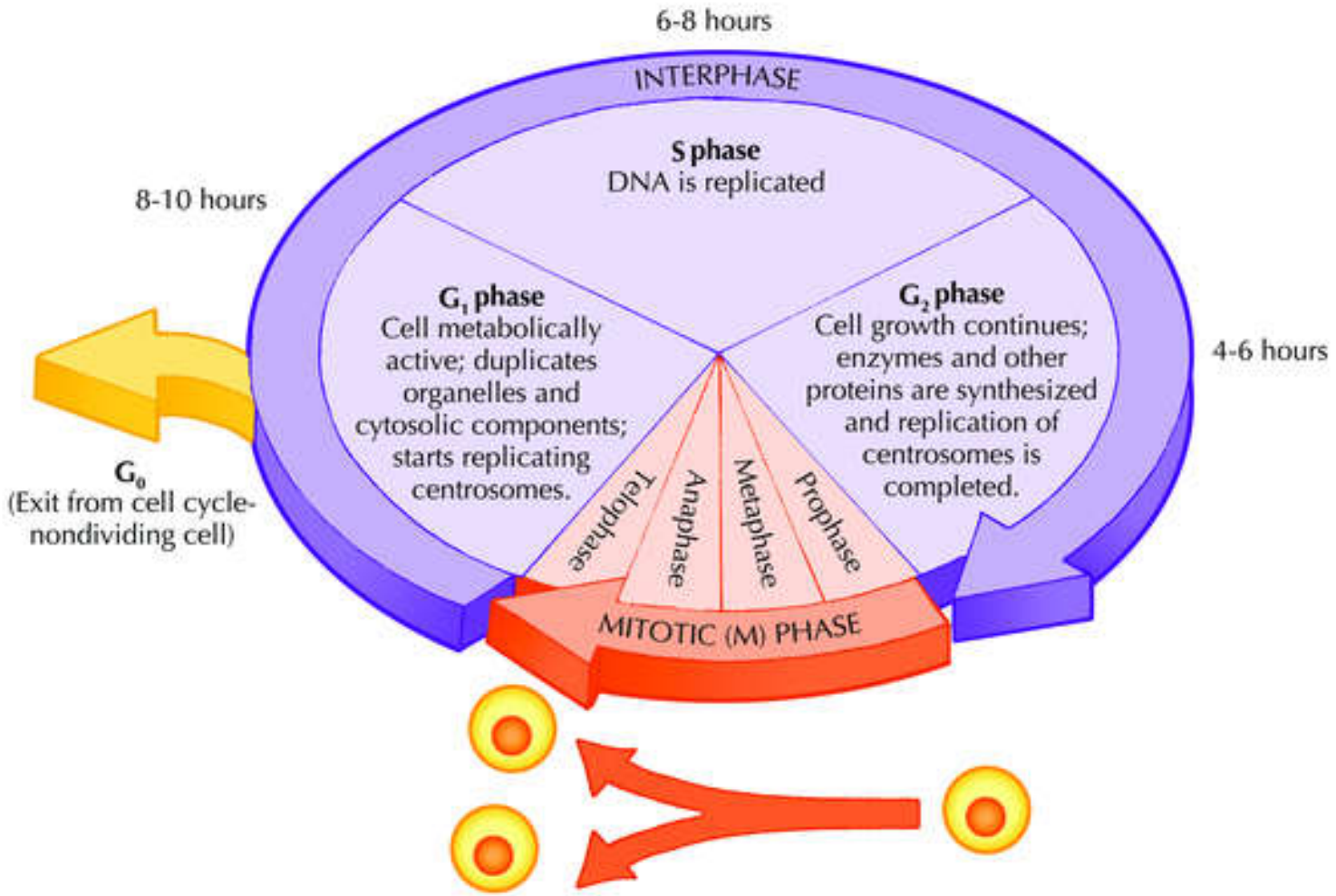
Cell division



Replication Facts

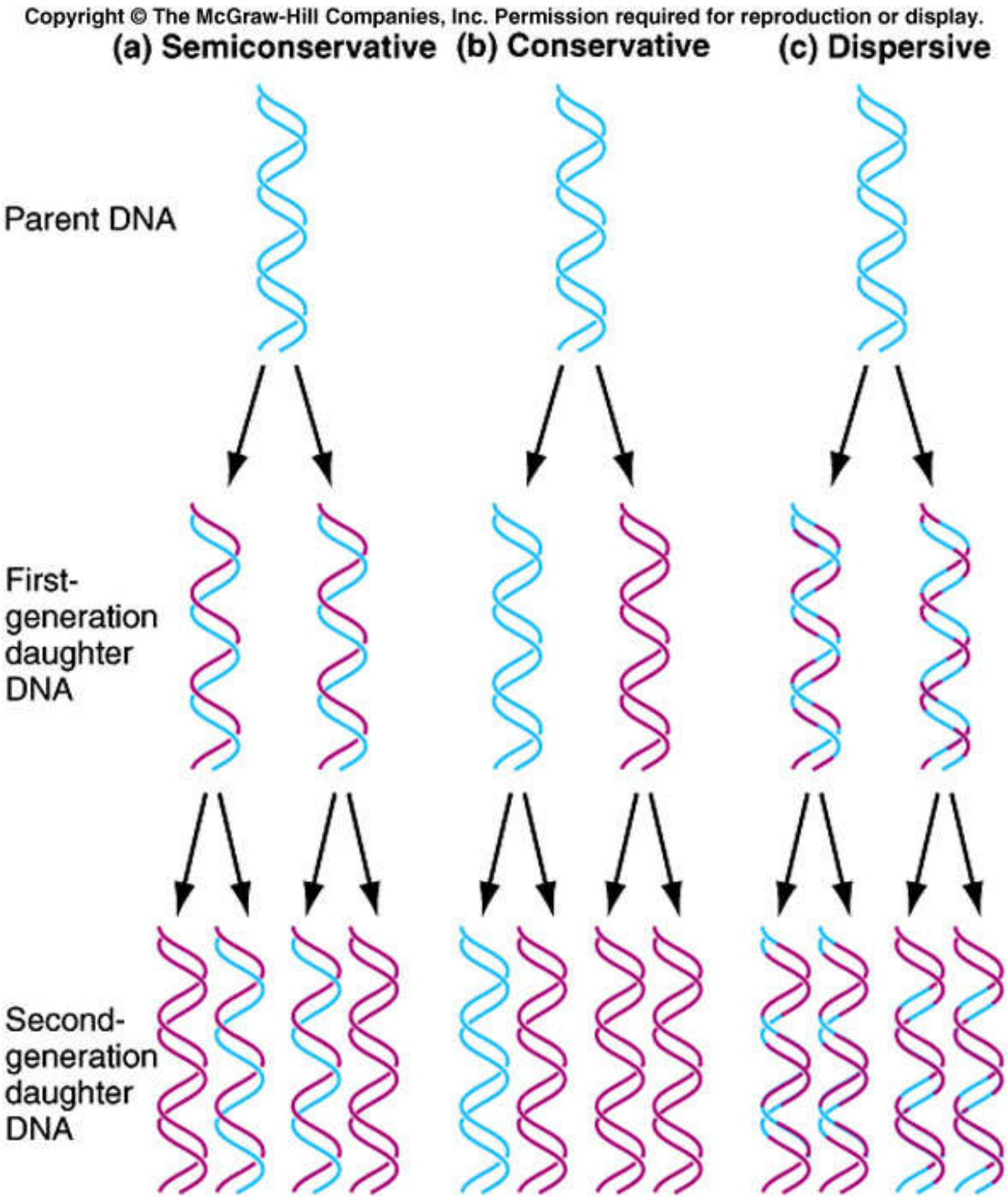
- DNA has to be copied before a cell divides
- DNA is copied during the S or synthesis phase of interphase
- New cells will need identical DNA strands

DNA is replicated during the Synthesis phase



DNA replication

3 possible models



Meselson and Stahl

1957

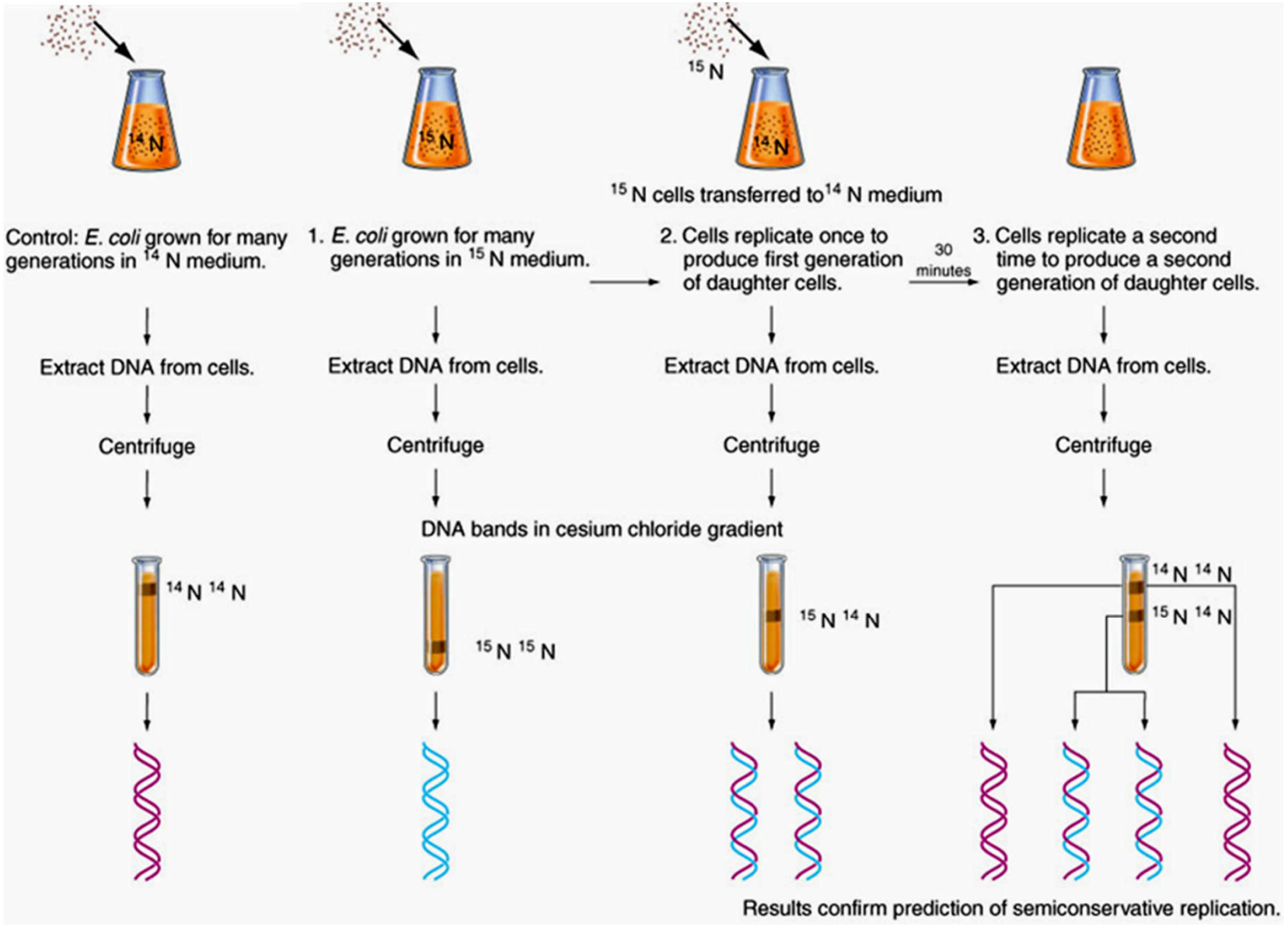
- Density Shift Experiments
- Labeled DNA with “heavy” Nitrogen
- Follow label:

Does it split equally = Semi-conservative

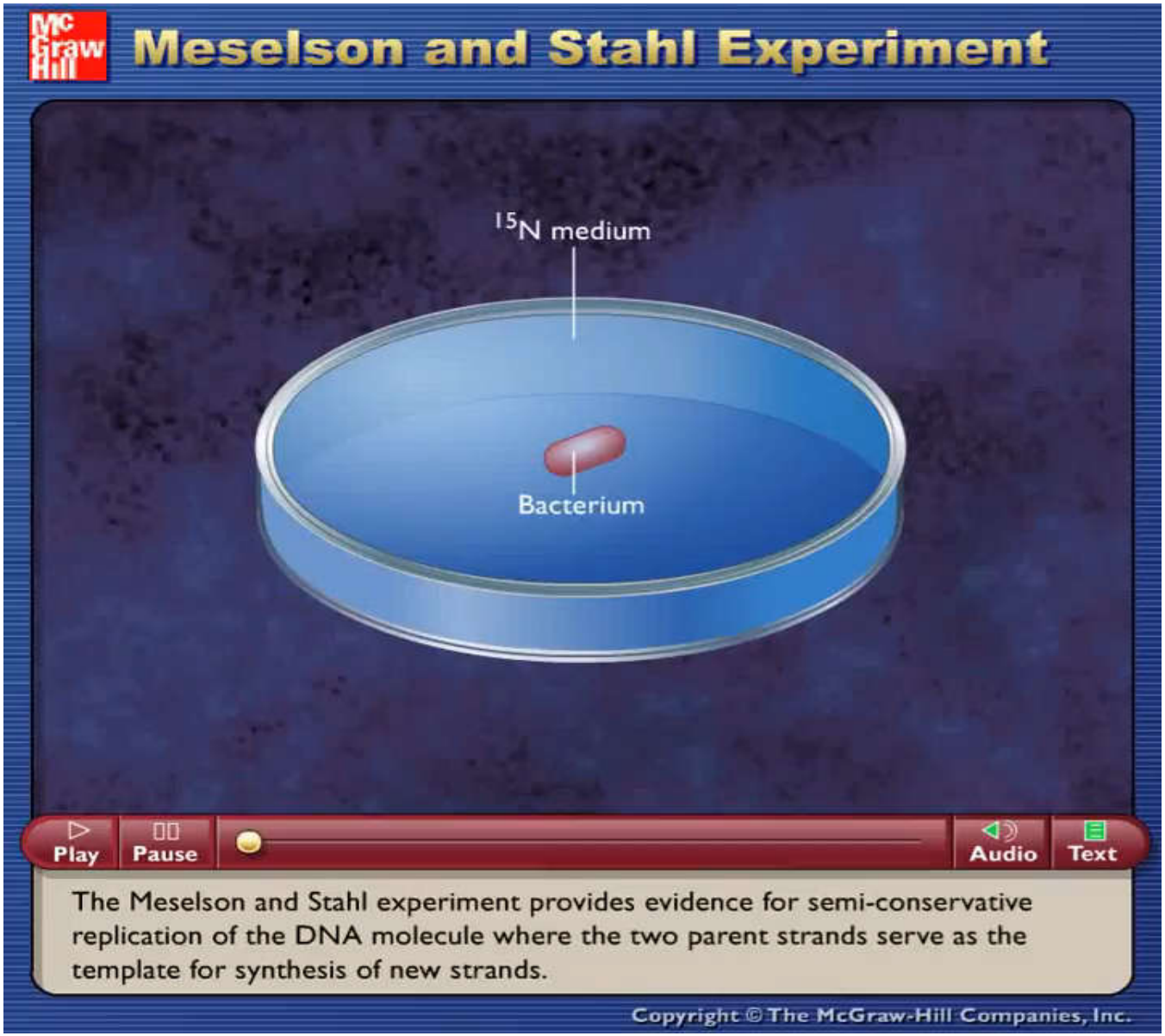
Does it stay together = Conservative

Does it split randomly = Dispersive

How do we know that DNA replication is semiconservative?
Meselson-Stahl experiments



Meselson-Stahl experiments

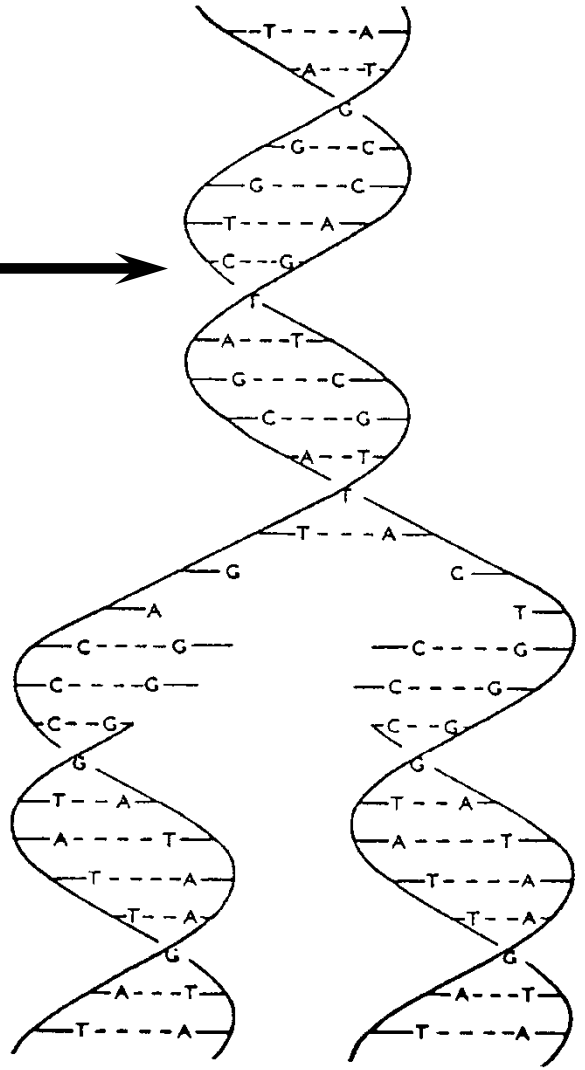


Semi-conservative

Parental DNA strands

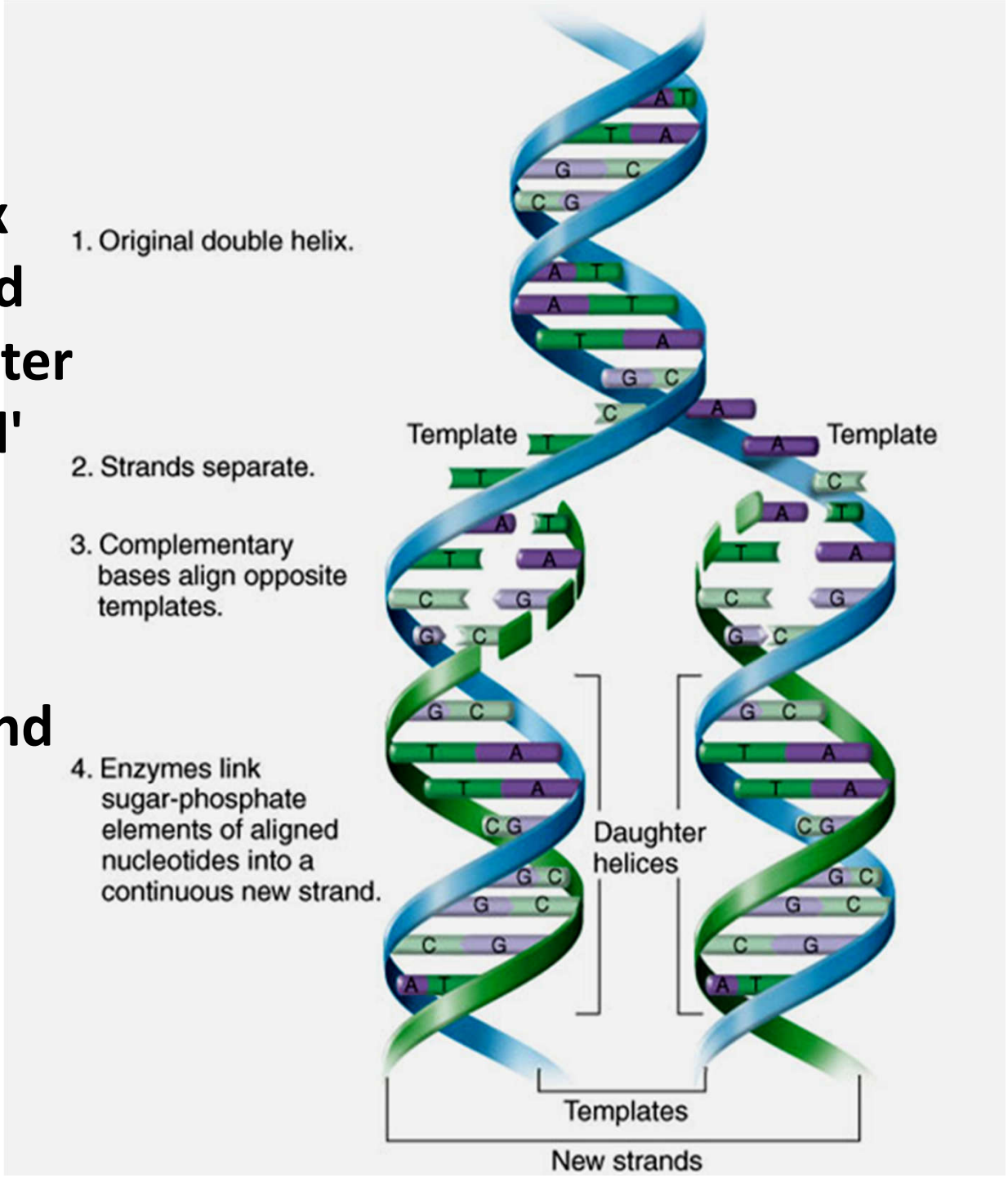
Each of the parental strands serves as a template for a daughter strand

Daughter DNA strands



Semi-conservative replication:

One strand of duplex passed on unchanged to each of the daughter cells. This 'conserved' strand acts as a template for the synthesis of a new, complementary strand by the enzyme DNA polymerase



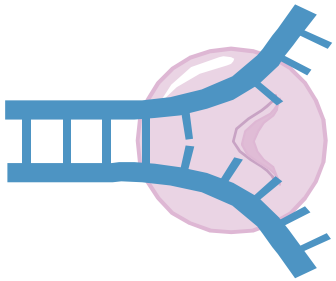
DNA Replication

- DNA from cell to cell, has to be replicated with fidelity
- Mistakes – mutations, cancer, etc.
- Only one mistake in a billion nucleotides is made by the replication machinery
- DNA replication is a result of the coordination of > 10 protein complexes and enzymes

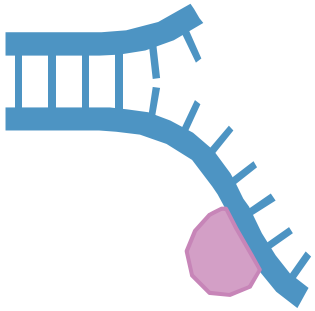
Enzymes

- Helicases – unwind the DNA
- Primase – attracts complimentary bases to form a “primer” sequence
- DNA Polymerase – add bases to the primer strand by reading the code
  - Therefore, extends the new strand
  - According to the original strand’s sequence
- Ligase – seals the sugar-phosphate backbone back together

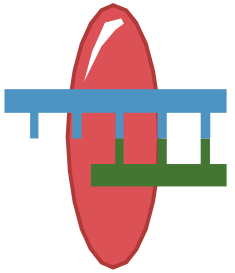
Enzymes in DNA replication



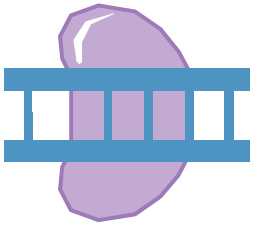
Helicase unwinds parental double helix



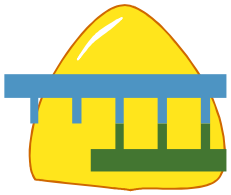
Binding proteins stabilize separate strands



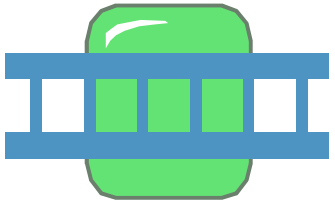
Primase adds short primer to template strand



DNA polymerase III binds nucleotides to form new strands



DNA polymerase I (Exonuclease) removes RNA primer and inserts the correct bases

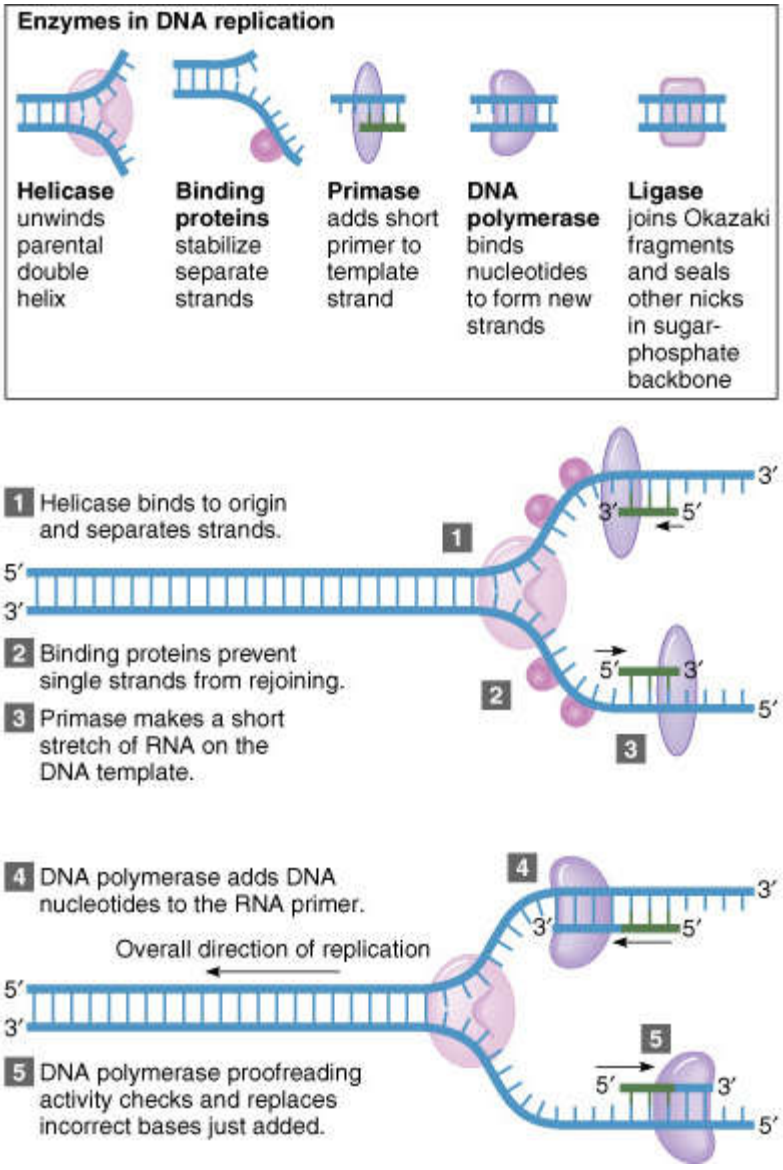


Ligase joins Okazaki fragments and seals other nicks in sugar-phosphate backbone

Synthesis

- Two strands
- Opposite directions
- One strand can be “read” directly because it is 5’ to 3’
  - Leading Strand
- Other strand is 3’ to 5’
  - Lagging Strand
  - Discontinuous synthesis of this strand

DNA Replication

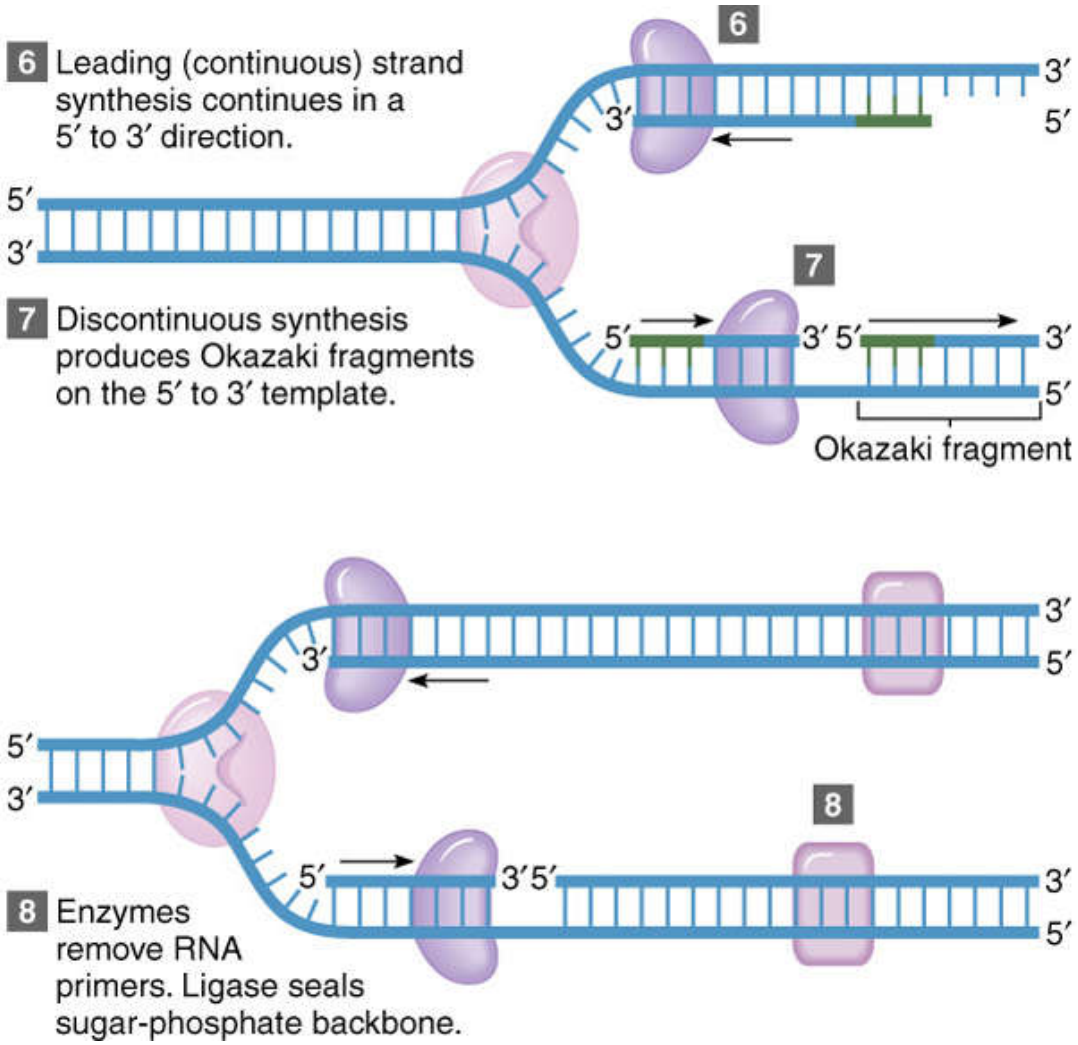


Notes:

1. Enzymes and their functions

2. Direction of replication
- Always 5’ to 3’

DNA Replication



Notes con't:

3. Difference between leading and lagging strands

Error Correction

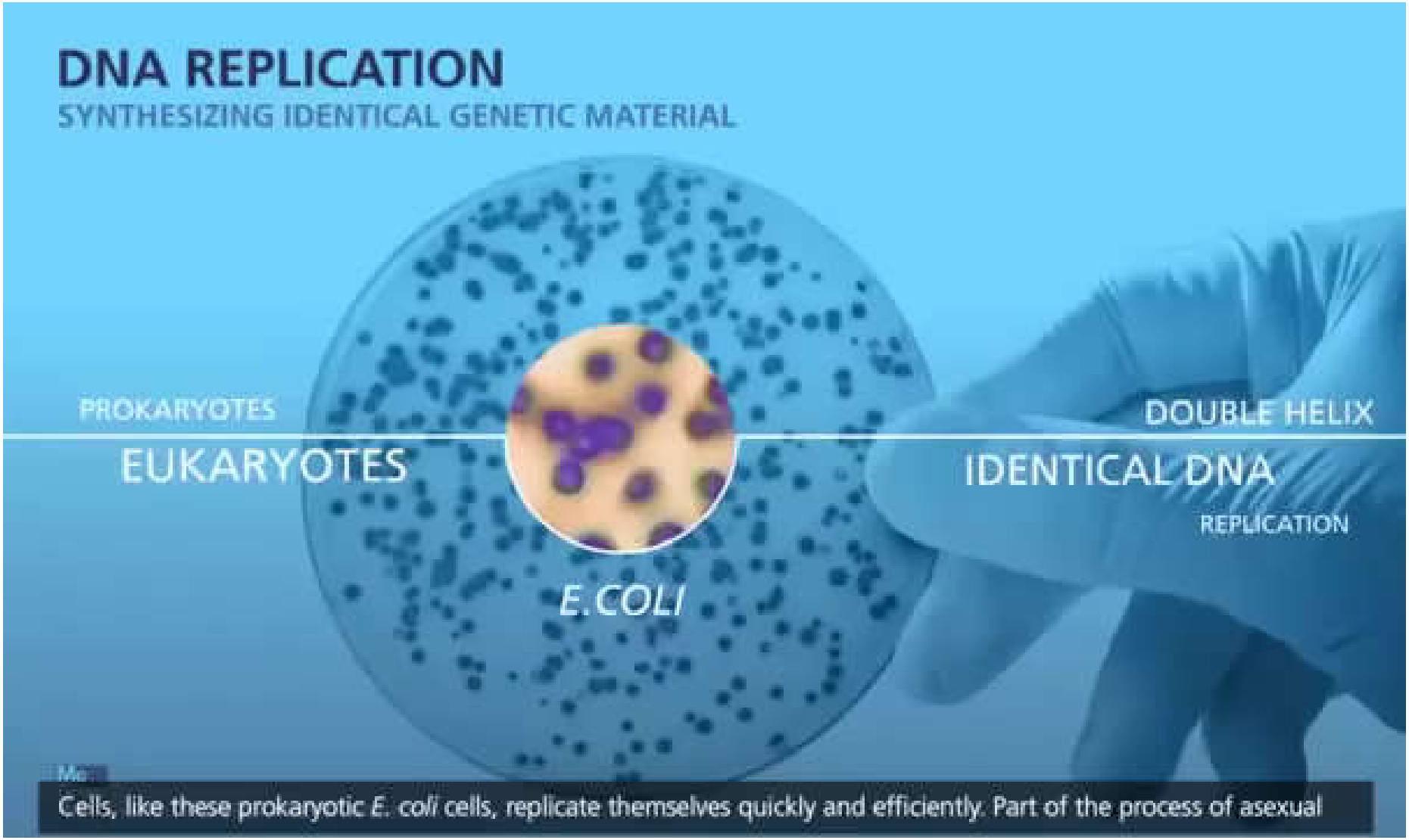
DNA Polymerase – can also "proofread" the newly formed strand:

1. Excise bases that are "mismatched"

2. Replace with the correct base

3. Then move forward to next base

DNA replication



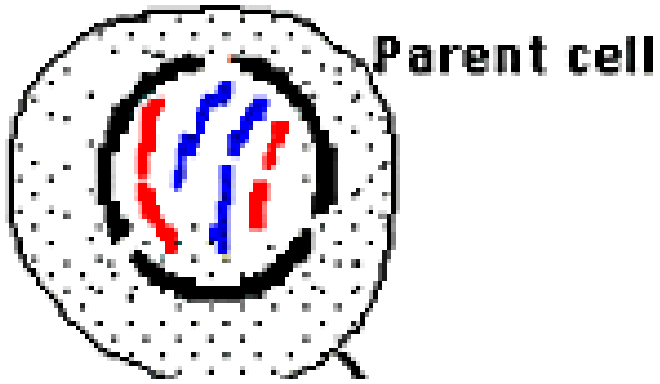
Mitosis

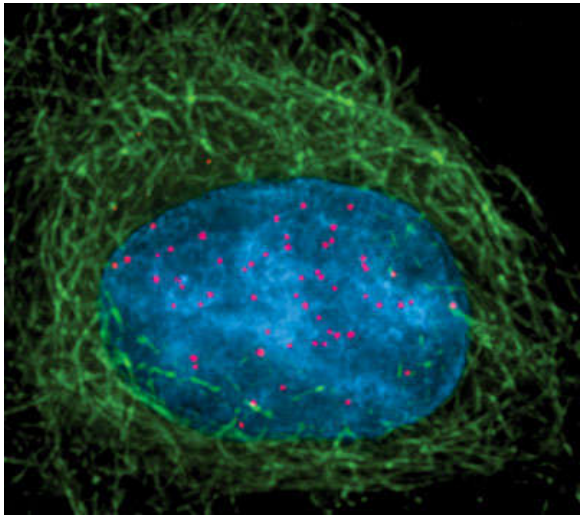
- To replace and repair other cells that have been damaged or worn out
- To allow multicellular organisms to grow and develop
- For asexual reproduction

Interphase

- **Interesting** things happen!

1. Cell preparing to divide
2. Genetic material doubles





- Interphase

Mitosis

- Prophase
- Metaphase
- Anaphase
- Telophase

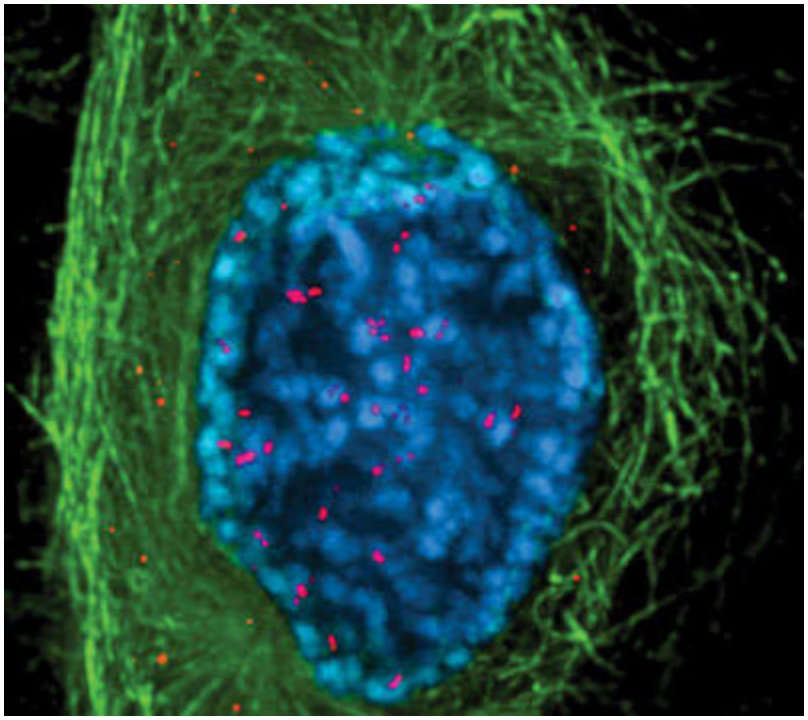
Prophase

- Chromosome **pair up**!

1. Chromosomes thicken and shorten

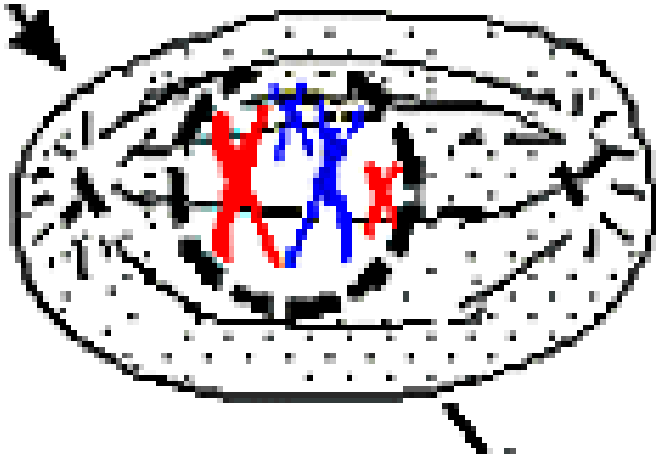
   -become visible

   -2 **chromatids** joined by a **centromere**
2. **Centrioles** move to the opposite sides of the nucleus
3. **Nucleus** disappears
4. Nuclear membrane disintegrate



microtubules

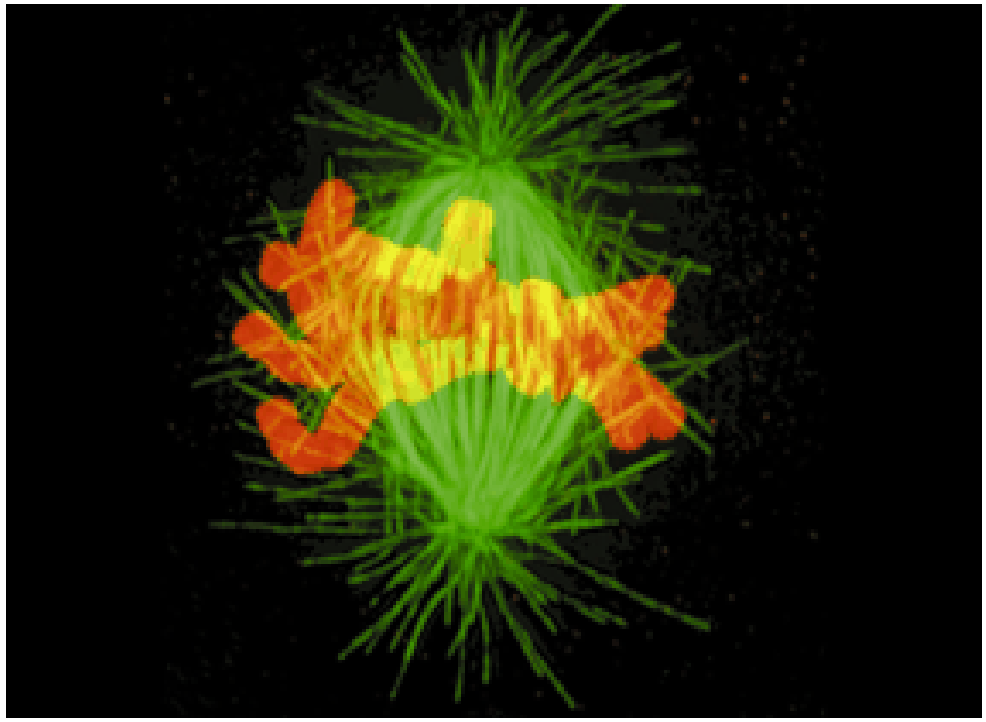
Prophase



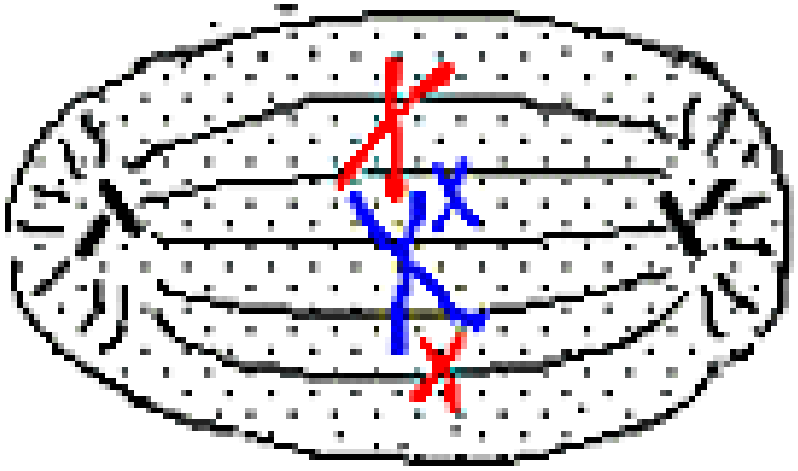
Metaphase

- Chromosomes **meet in the middle**!

1. Chromosomes arrange at **equator** of cell
2. Become attached to **spindle fibres** by **centromeres**
3. **Homologous** chromosomes **do not associate**



Metaphase

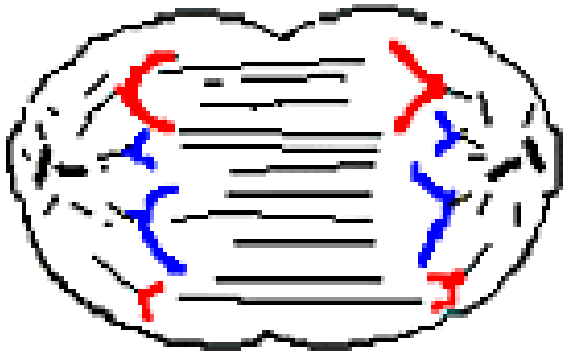


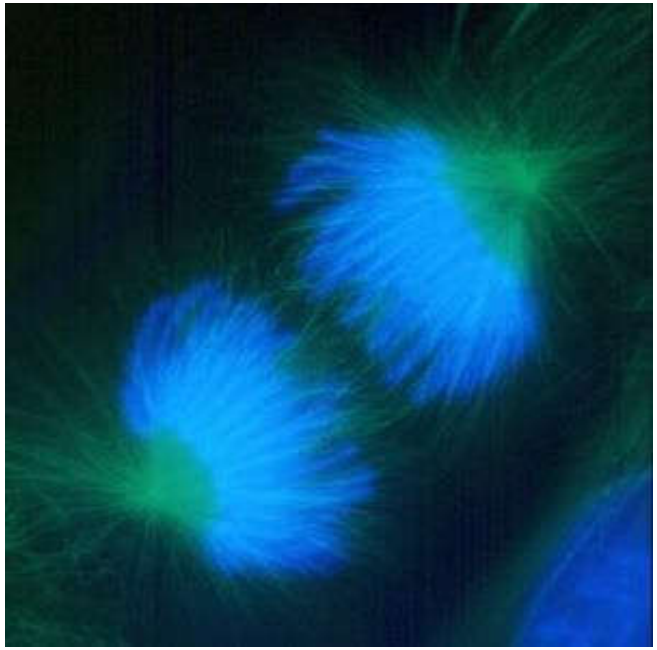
Anaphase

- Chromosomes get pulled **apart**

1. Spindle fibres contract pulling chromatids to the opposite poles of the cell

Anaphase

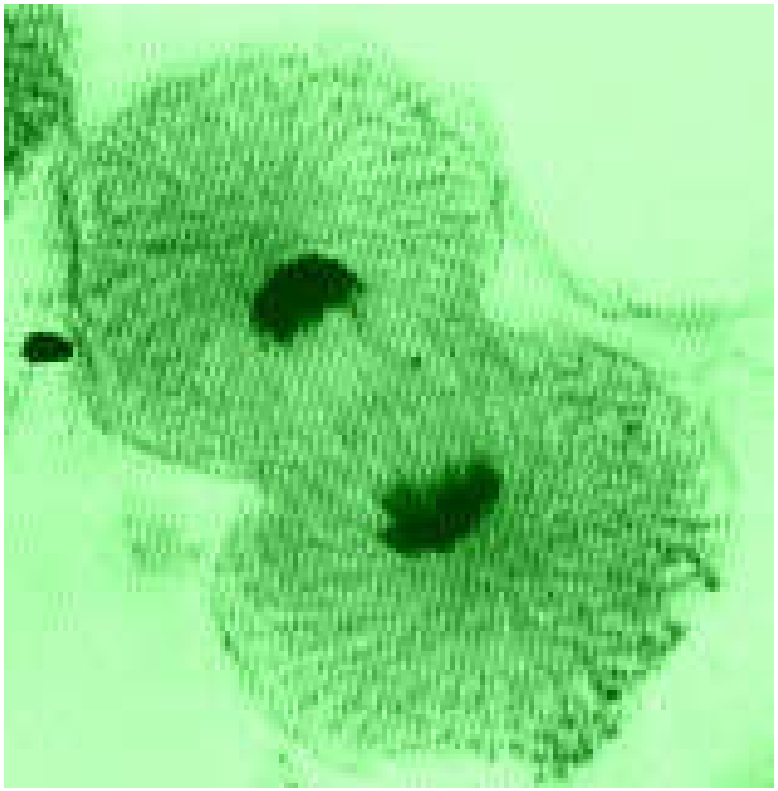




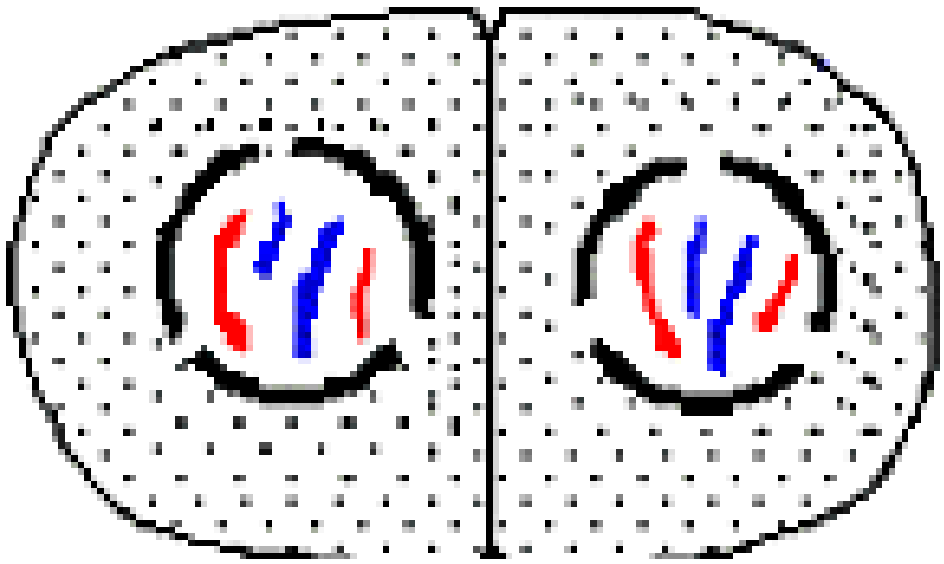
Telophase

- Now there are **two!**

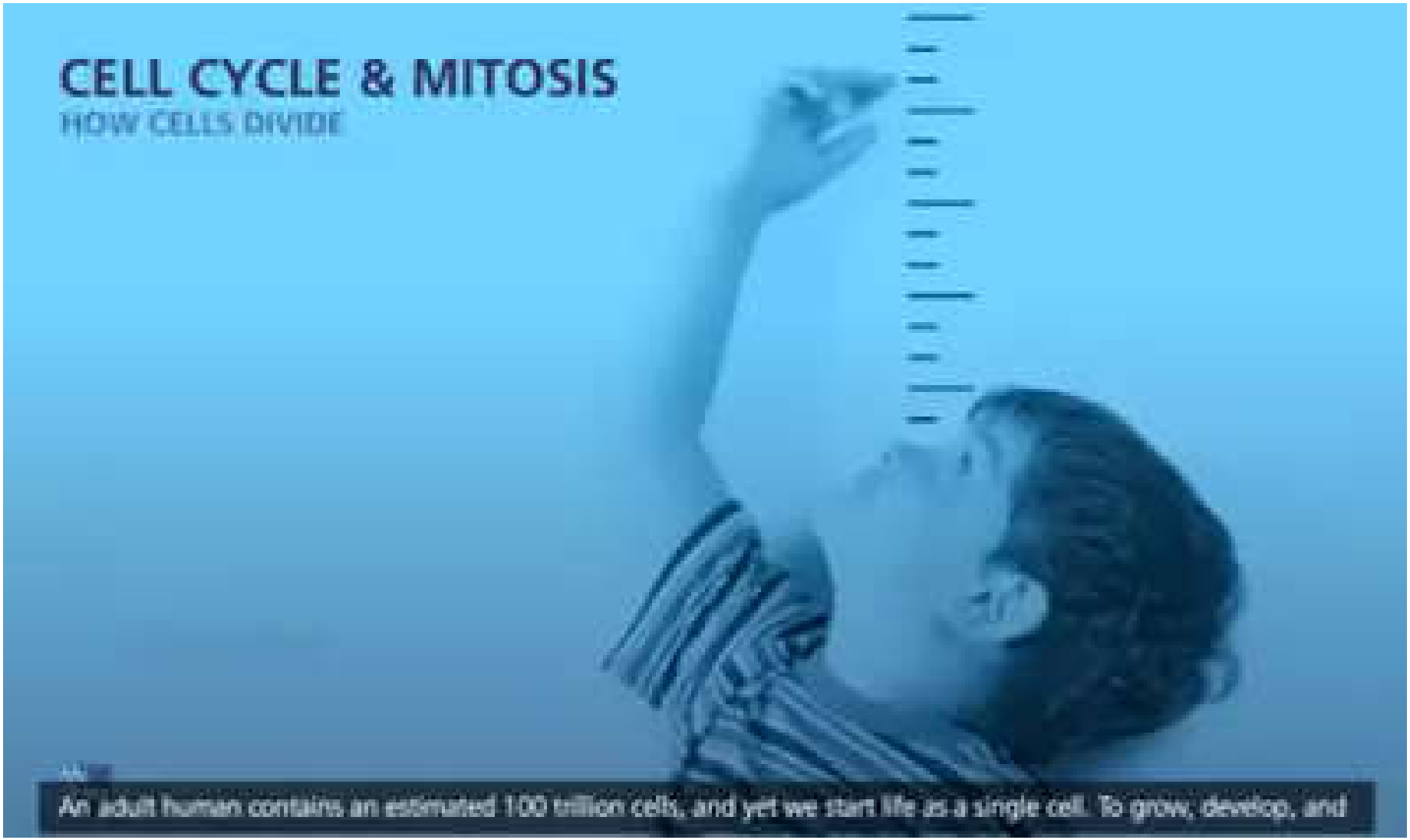
1. Chromosomes uncoil
2. Spindle fibres disintegrate
3. Centrioles replicate
4. Nucleur membrane forms
5. Cell divides



Telophase

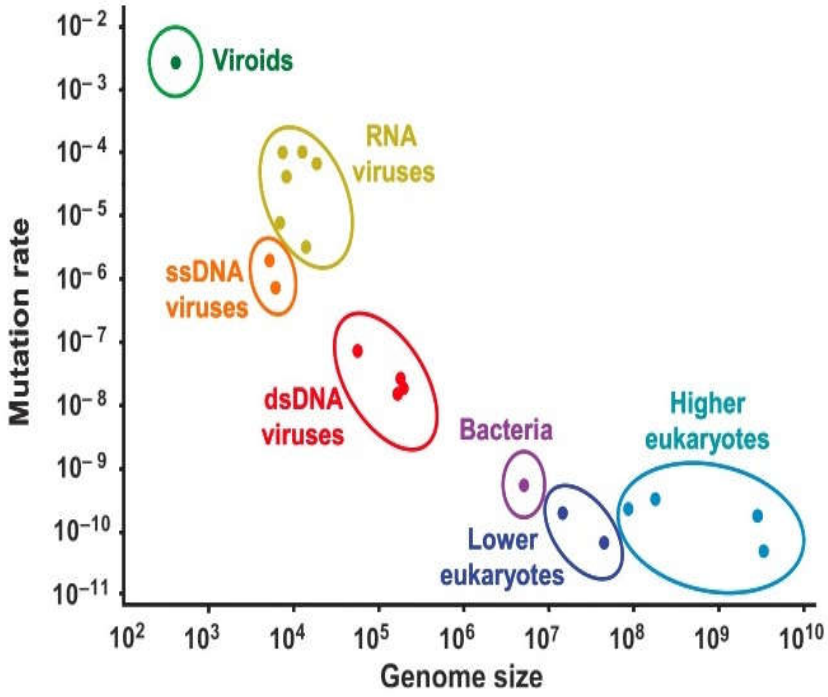


Mitosis


CELL CYCLE & MITOSIS
HOW CELLS DIVIDE
An adult human contains an estimated 100 trillion cells, and yet we start life as a single cell. To grow, develop, and

MUTATION RATE

- DNA viruses have mutation rates between 10^{-6} to 10^{-8} mutations per base per generation.
- RNA viruses have mutation rates between 10^{-3} to 10^{-5} per base per generation.
- In general, the mutation rate in unicellular eukaryotes and bacteria is roughly 0.003 mutations per genome per *cell* generation.
- Human mitochondrial DNA has been estimated to have mutation rates of ~3× or ~2.7×10^{-5} per base per 20 year generation.
- The human genome mutation rate is similarly estimated to be ~1.1×10^{-8} per site per generation.
- The human mutation rate is higher in the male germ line (sperm) than the female (egg cells)



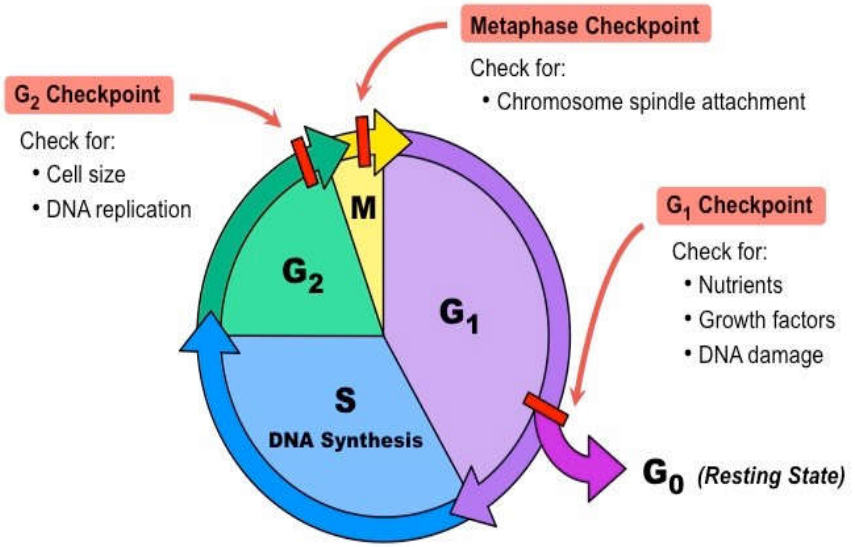
CELL CYCLE CHECKPOINT

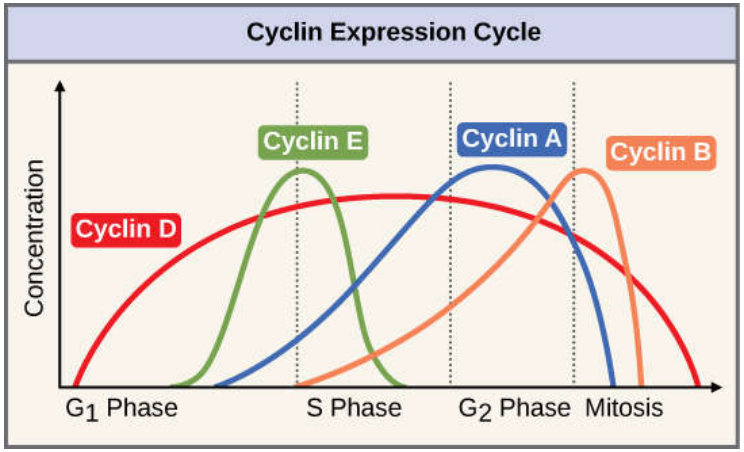
- The **G1 checkpoint (G1/S checkpoint)** is the main decision point for a cell – that is, the primary point at which it must choose whether or not to divide. Once the cell passes the G1 checkpoint and enters S phase, it becomes irreversibly committed to division.

- Monitors the cell for DNA damage or other problems before the cell enters the S phase and replicates its DNA.

- This checkpoint is controlled by the tumor suppressor **protein p53**, which can activate the expression of genes that pause the cell cycle and allow time for DNA repair to occur. Other proteins, such as cyclin-dependent kinase inhibitors (**CDKIs**) and **cyclin D, E** , also play a role in regulating the G1/S checkpoint.

Cyclin D – CDK4, CDK6 -> active Rb -> release E2F
Activate cyclin E – CDK2

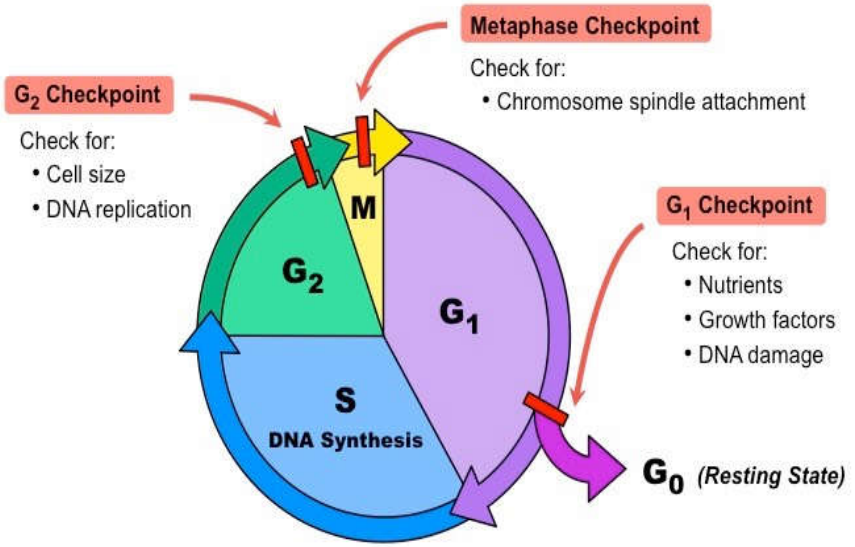


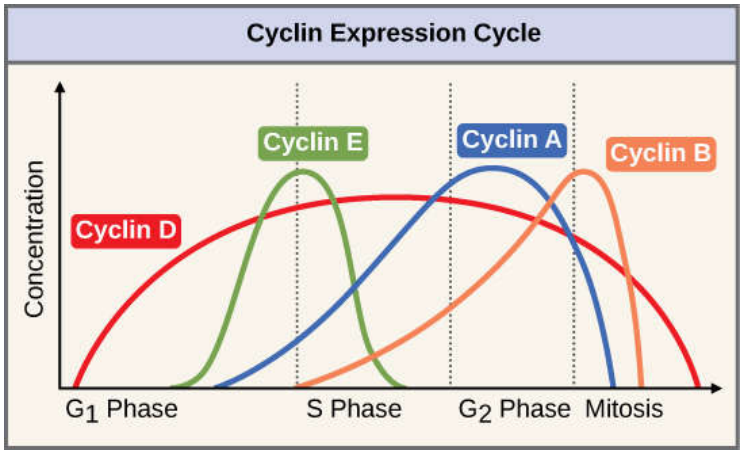


CELL CYCLE CHECKPOINT

- The **G2 checkpoint** (**G2/M checkpoint**) make sure that cell division goes smoothly, the cell has an additional checkpoint before M phase, called. If errors or damage are detected, the cell will pause at the G2 checkpoint to allow for repairs.

- The G2/M checkpoint monitors the cell after DNA replication has occurred, but before the cell enters mitosis. This checkpoint is regulated by concentration of **Cyclin A**, **the protein kinase CDK1** and **CDK2**, which can activate other proteins that delay the cell cycle and allow time for DNA repair or completion of replication.

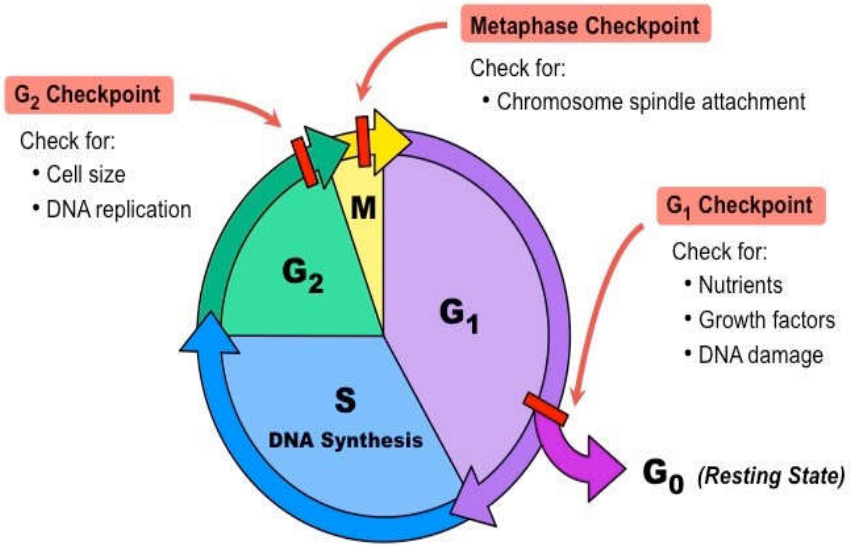


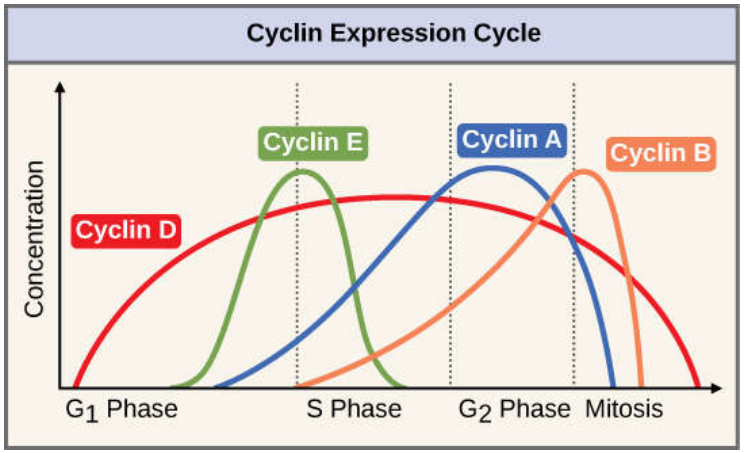


CELL CYCLE CHECKPOINT

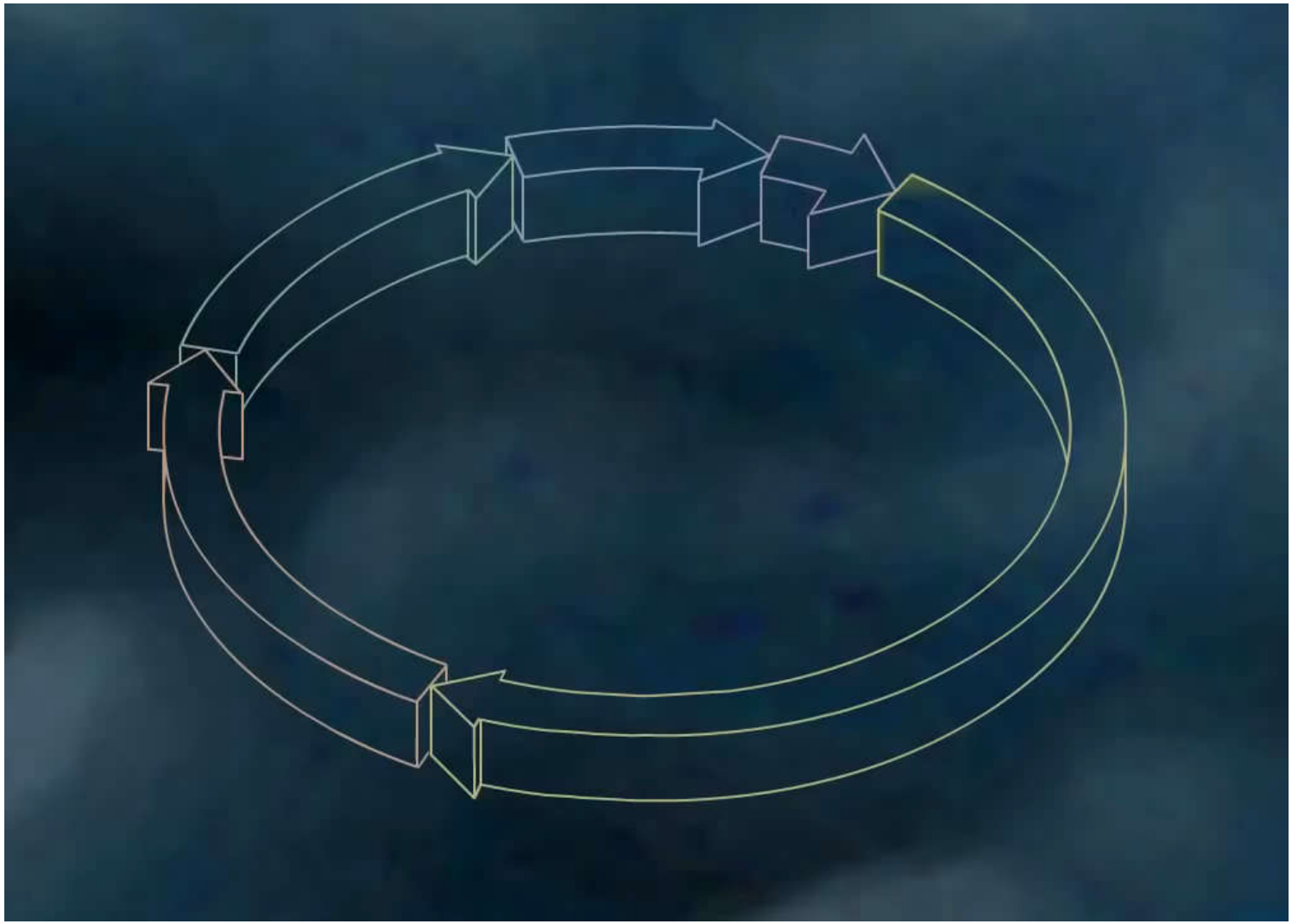
- The **M checkpoint** is also known as the **spindle checkpoint**: here, the cell examines whether all the sister chromatids are correctly attached to the spindle microtubules (Prophase of M phase)

- The spindle checkpoint monitors the proper attachment of chromosomes to the mitotic spindle during mitosis. This checkpoint is regulated by several proteins, including **cyclin B – CDK1**, which can delay the onset of anaphase until all chromosomes are properly aligned and attached to the spindle.





CELL CYCLE CHECKPOINT



CELL CYCLE

- In human cells, the S, G2, and the M phases together usually take about 12 hours
- Length of G1 phase differs most for different cell types
- Rate of cell division greater in embryos and children

CELL TYPE	APPROXIMATELY LIFE SPAN
Skin cell	2 weeks
Red blood cell	4 months
Liver cell	300-500 days
Intestine	4-5 days
Muscle	16 years

Rate of cell division

Four Basic Stages of Cancer



Cell with genetic mutation

Cancer cells Microscopic

A

Cancer is large enough to be felt as a lump

B

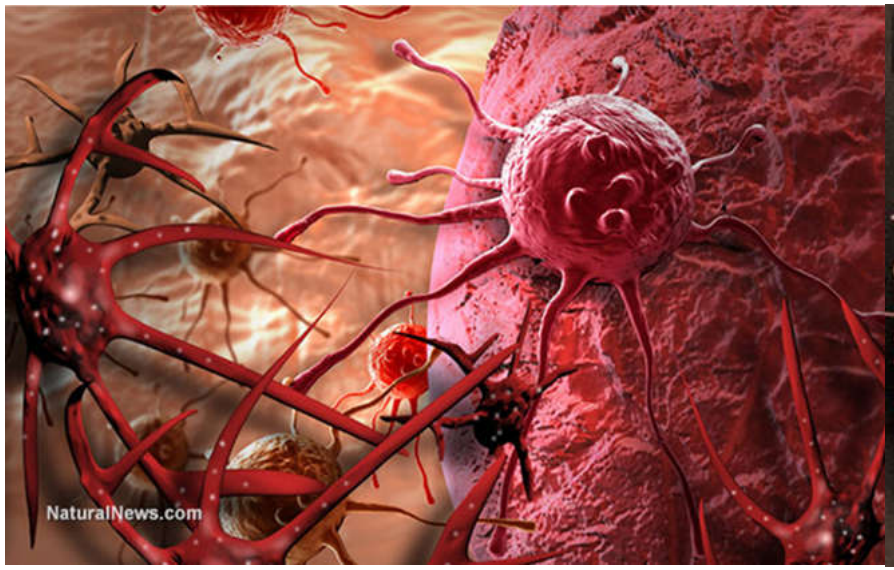
Cancer has spread from the original site into surrounding areas

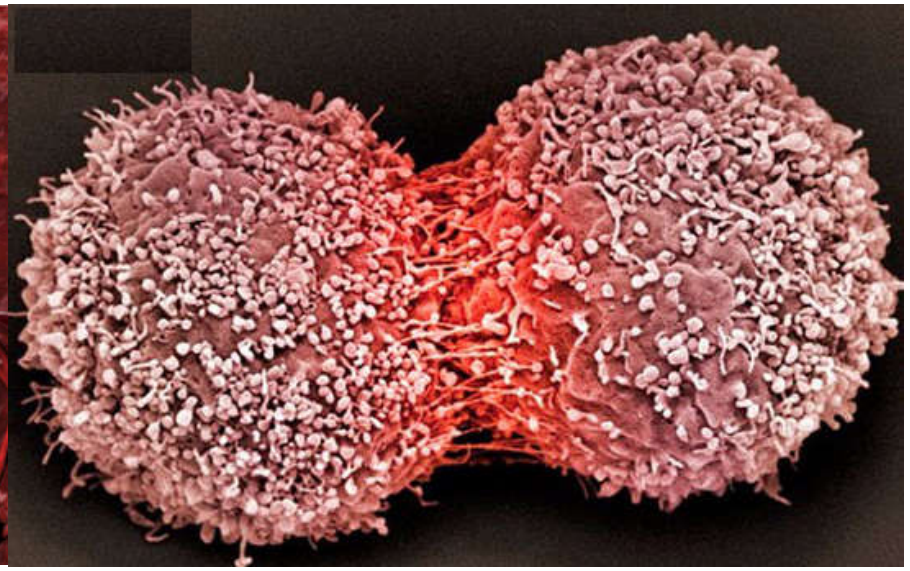
C

Cancer has spread to the other tissue parts of the body

D

WE ALL HAVE CANCER CELL IN OUR BODIES





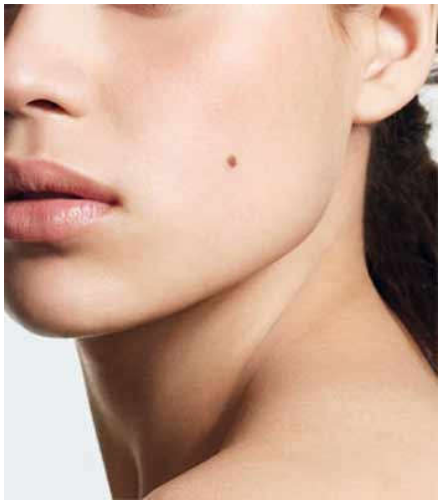
Science News *from research organizations*

Cancer Cells Are Protected By Our Own Immune System

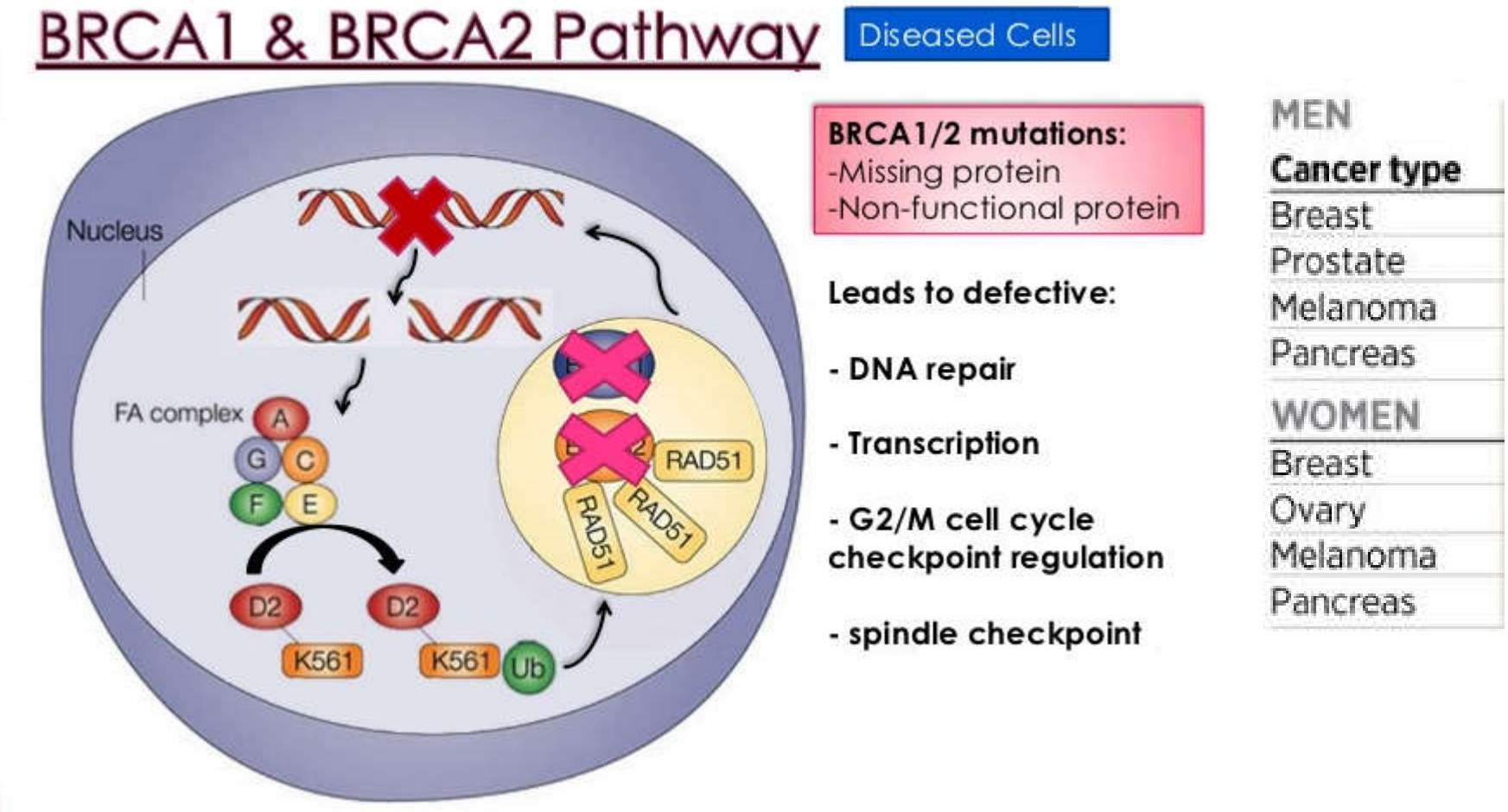
Date: August 10, 2009

Source: CNRS (Délégation Paris Michel-Ange)

Summary: During the very first few days of development of a cancer, our immune system recognizes cancer cells not as abnormal cells requiring eradication but as cells of the body that need to be protected, new research reveals.



CANCER RELATED GENES



Source: *Nature.com*

Questions

The G_1 phase of the cell cycle primarily involves?

1. Prepare DNA for replication.
2. DNA replication
3. Prepare for mitosis

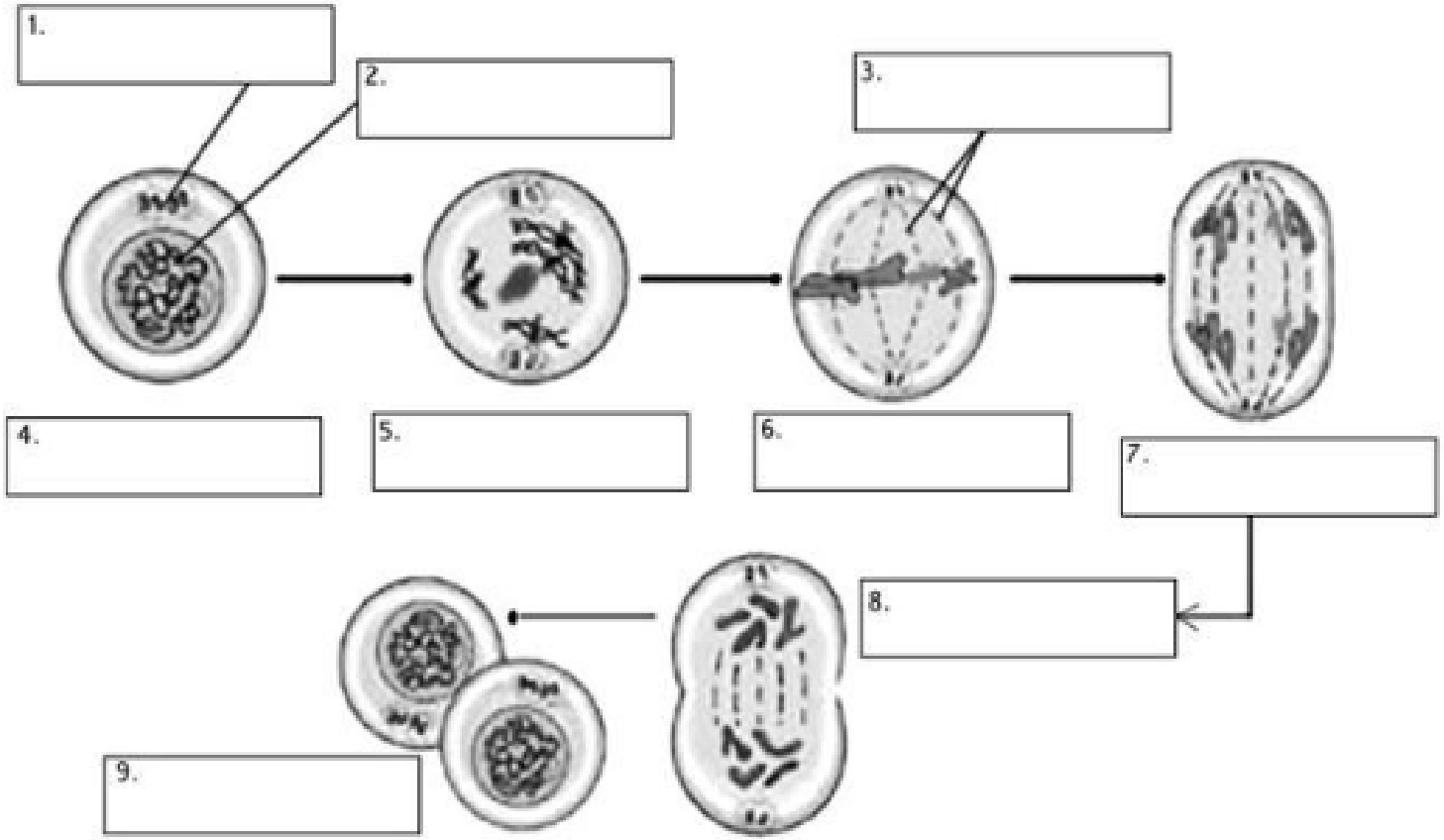
Questions

The G_1 phase of the cell cycle primarily involves?

1. Prepare DNA for replication.
2. DNA replication. (Phase S)
3. Prepare for mitosis. (Interphase)

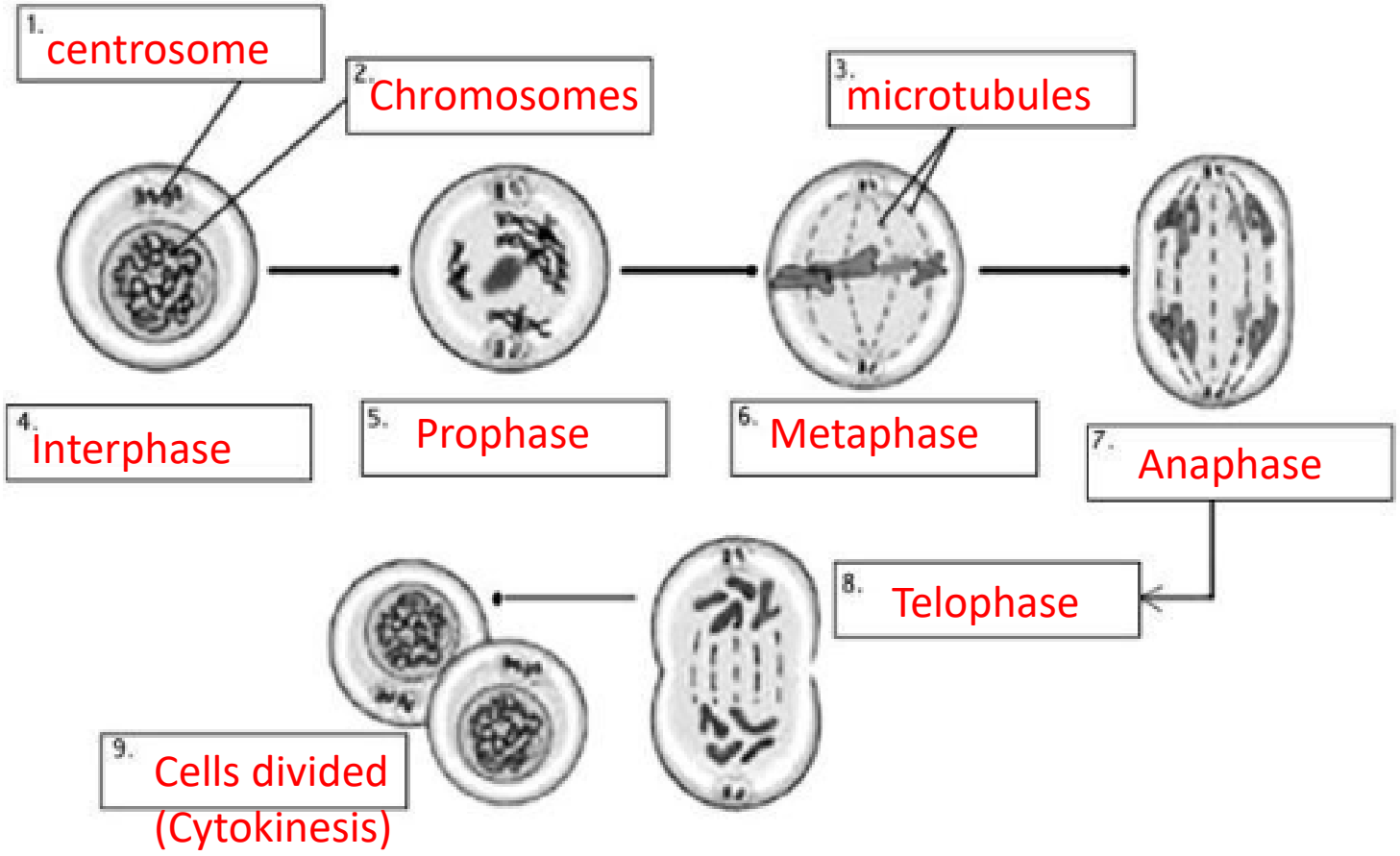
Questions

Mitosis practice exercise



Questions

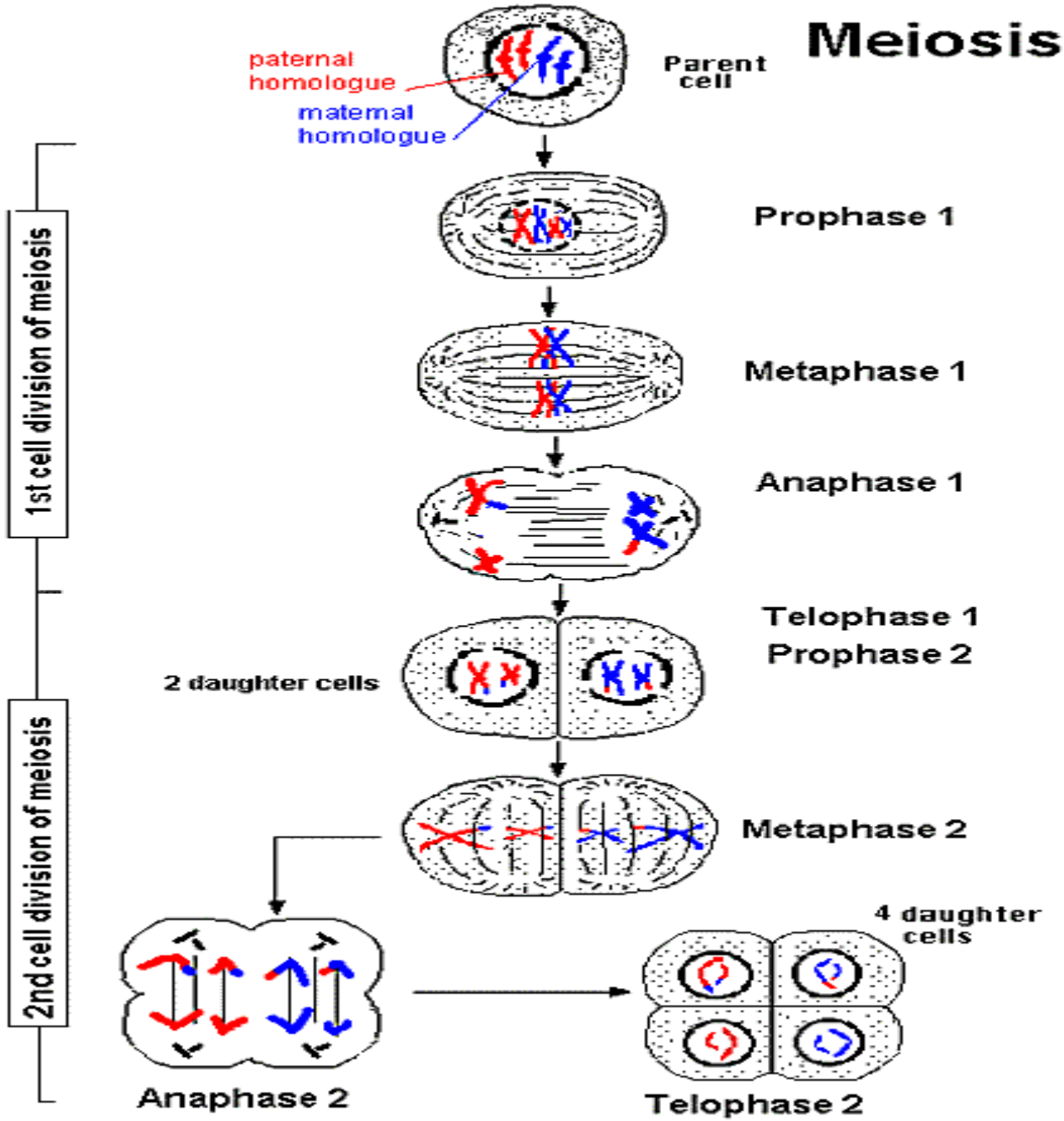
Mitosis practice exercise



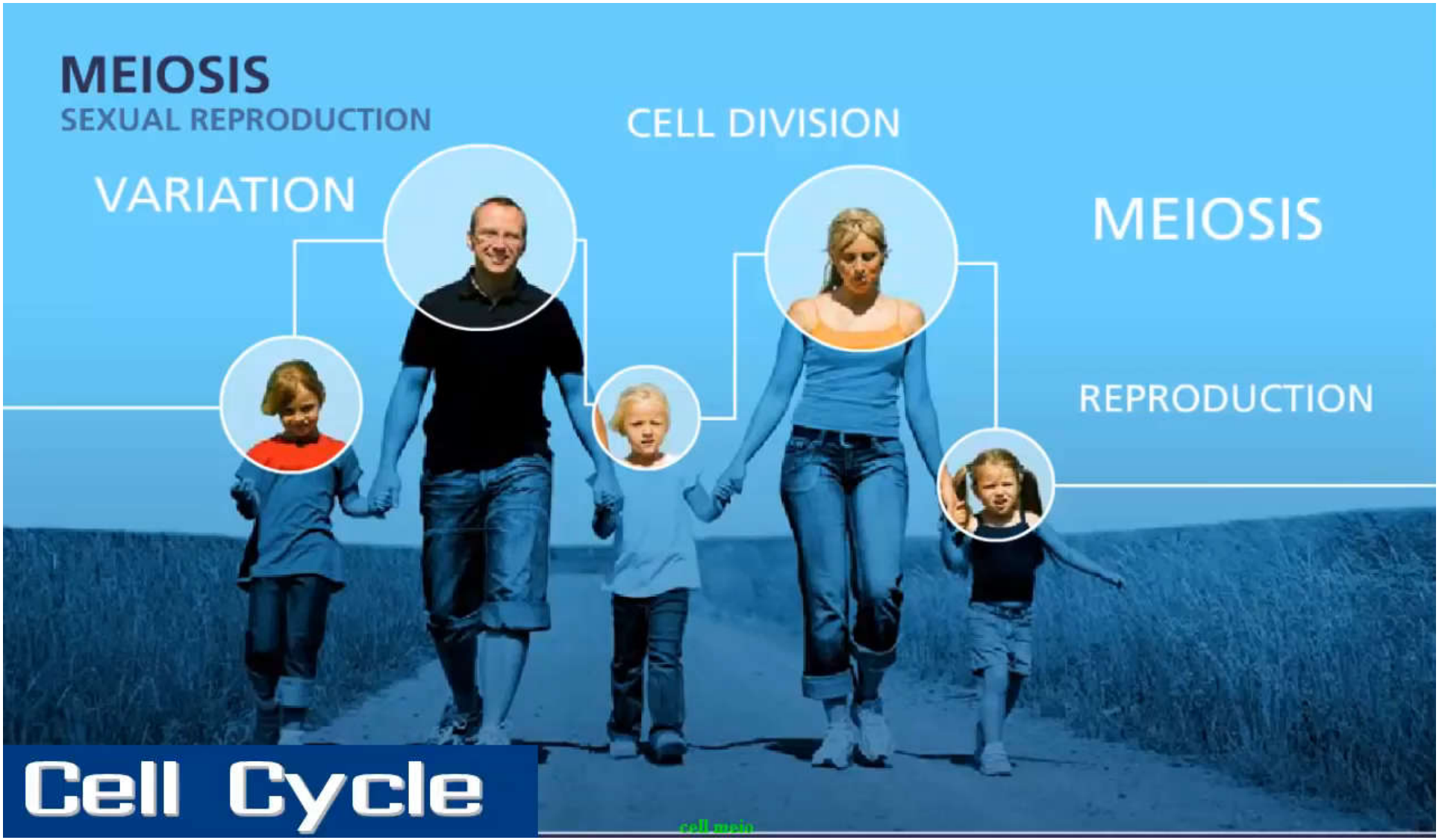
Meiosis

- Meiosis results in four cells with half the number of chromosomes so that when the sex cells (sperm and egg) combine, the original or normal number of chromosomes will be restored

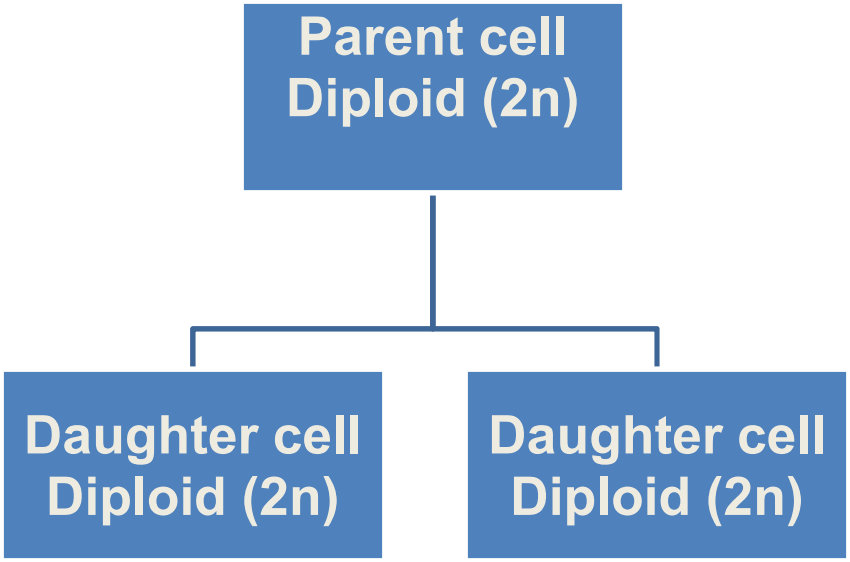
- 4 daughter cells produced
- Each daughter cell has half the chromosomes of the parent
- 2 sets of cell division involved


Meiosis
paternal homologue
maternal homologue
Parent cell
Prophase 1
Metaphase 1
Anaphase 1
Telophase 1
Prophase 2
2 daughter cells
Metaphase 2
4 daughter cells
Anaphase 2
Telophase 2
1st cell division of meiosis
2nd cell division of meiosis

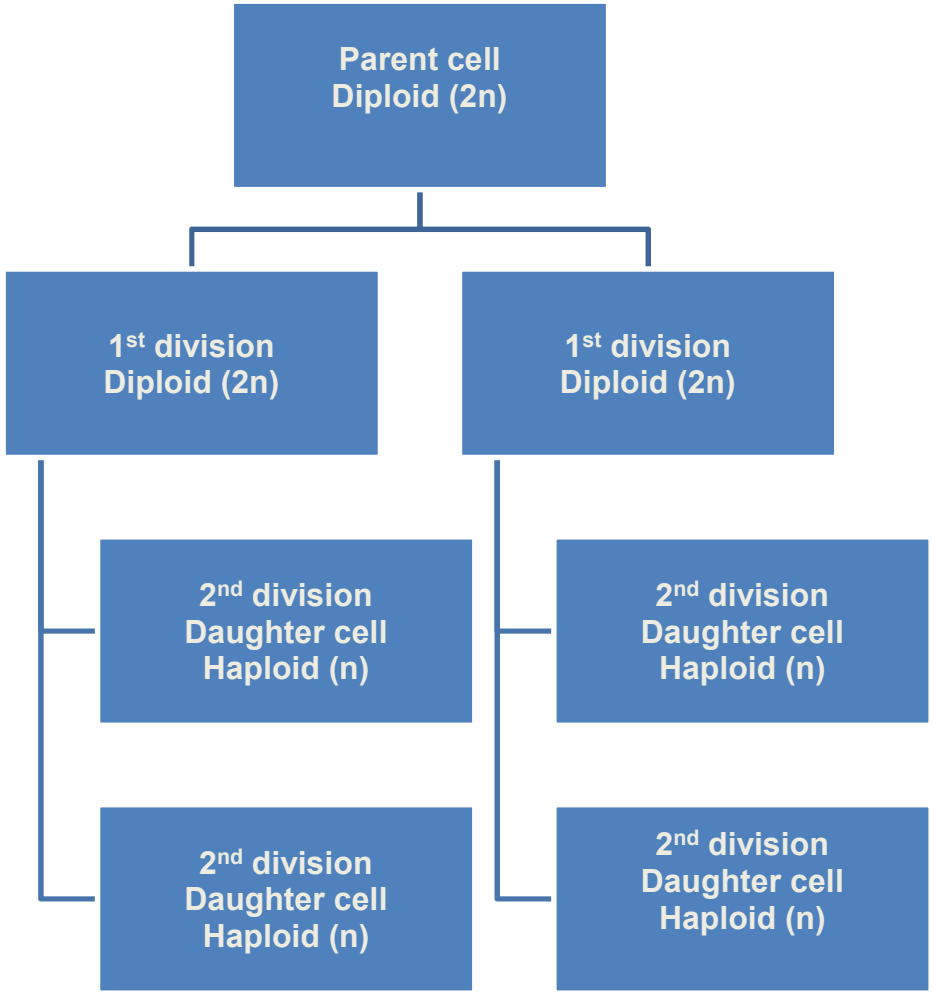
Meiosis


MEIOSIS
SEXUAL REPRODUCTION
VARIATION
CELL DIVISION
MEIOSIS
REPRODUCTION
Cell Cycle

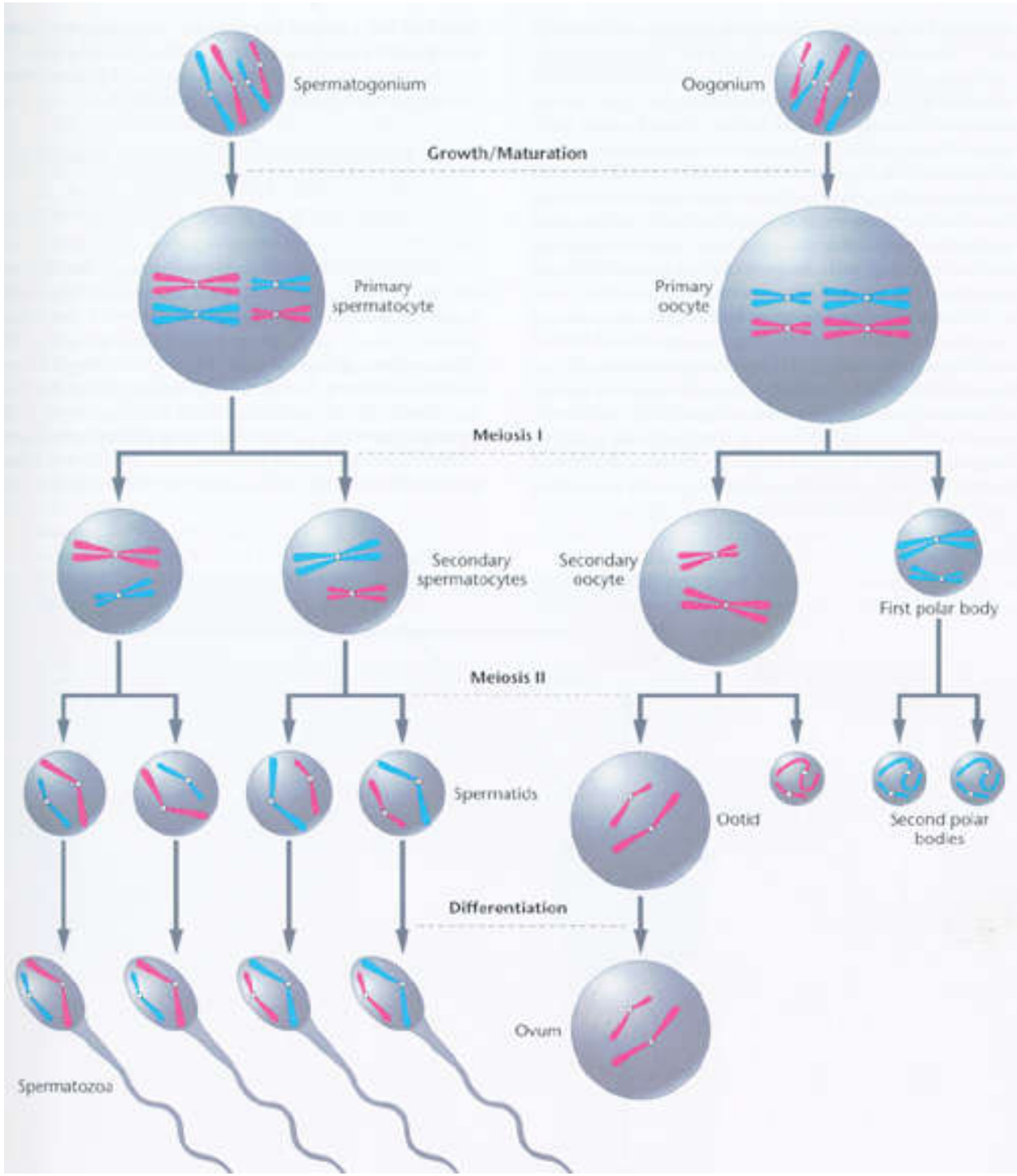
Mitosis



Meiosis



The difference between mitosis and meiosis is that mitosis produces two identical daughter cells and meiosis produces four genetically different daughter cells


Spermatogonium
Oogonium
Growth/Maturation
Primary spermatocyte
Primary oocyte
Meiosis I
Secondary spermatocytes
Secondary oocyte
First polar body
Meiosis II
Spermatids
Ootid
Second polar bodies
Differentiation
Ovum
Spermatozoa

MITOSIS

- 1 Cell Division
- Produces Soma Cells and primitive sex cells.
- From 1 parent cell produces 2 daughter cells
- Daughter Cells are Diploid **2N**
- Daughter cells **IDENTICAL** to parent cells

MEIOSIS

- 2 Cell Divisions
- Produces Sex Cells
- From 1 parent cell produces 4 daughter cells
- Daughter cells are haploid **N**
- Daughter cells are GENETICALLY different from parent cell due to **CROSSING OVER** of chromosomes

Questions

Select **all** true statements about meiosis.?

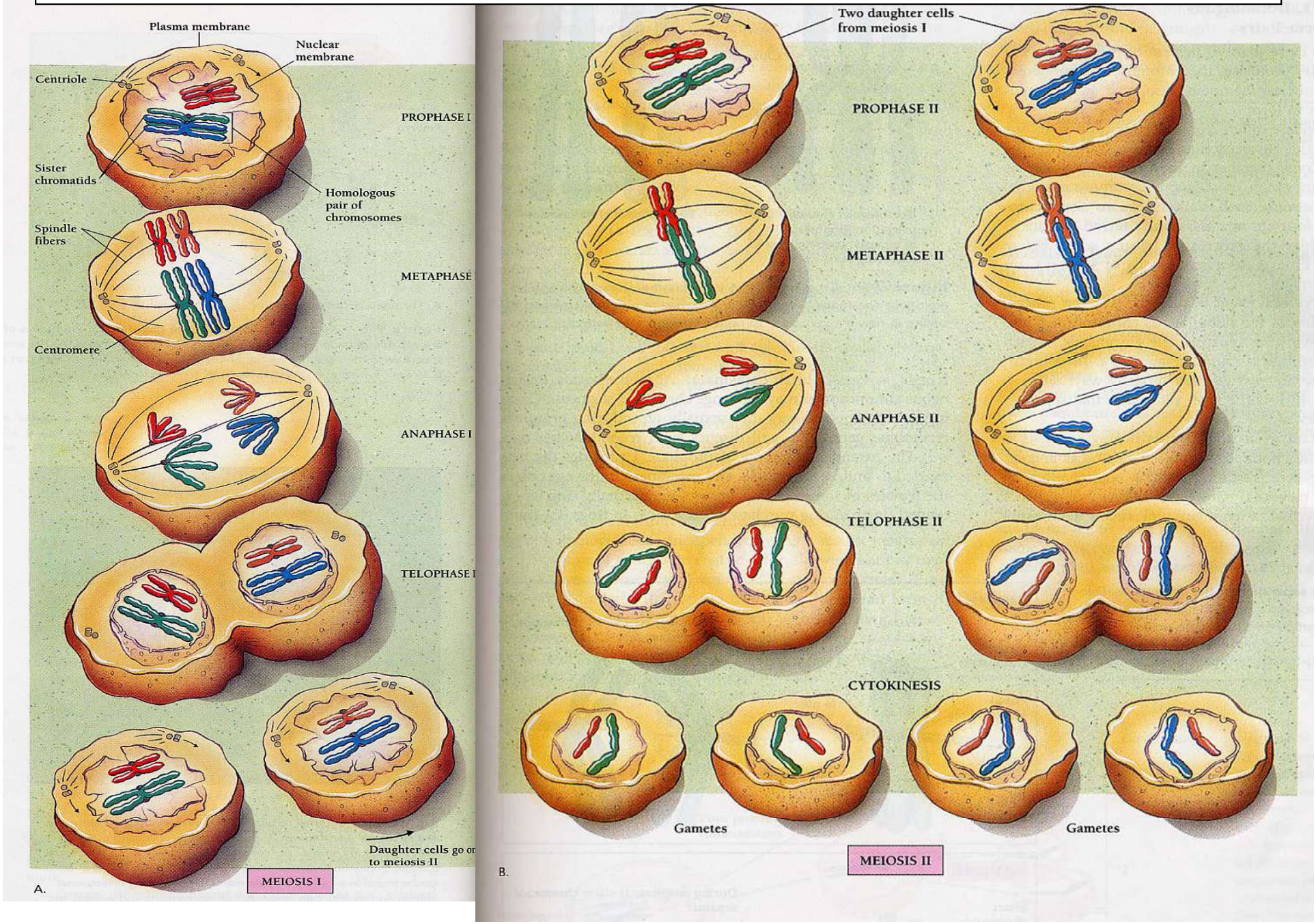
1. Meiosis involves two sequential cell divisions.
2. No DNA replication occurs before meiosis because the chromosome set is halved.
3. Crossing-over leads to variation in the genetic material.
4. During Meiosis I, paternal and maternal chromsomes are separated, ending up int two different cells.

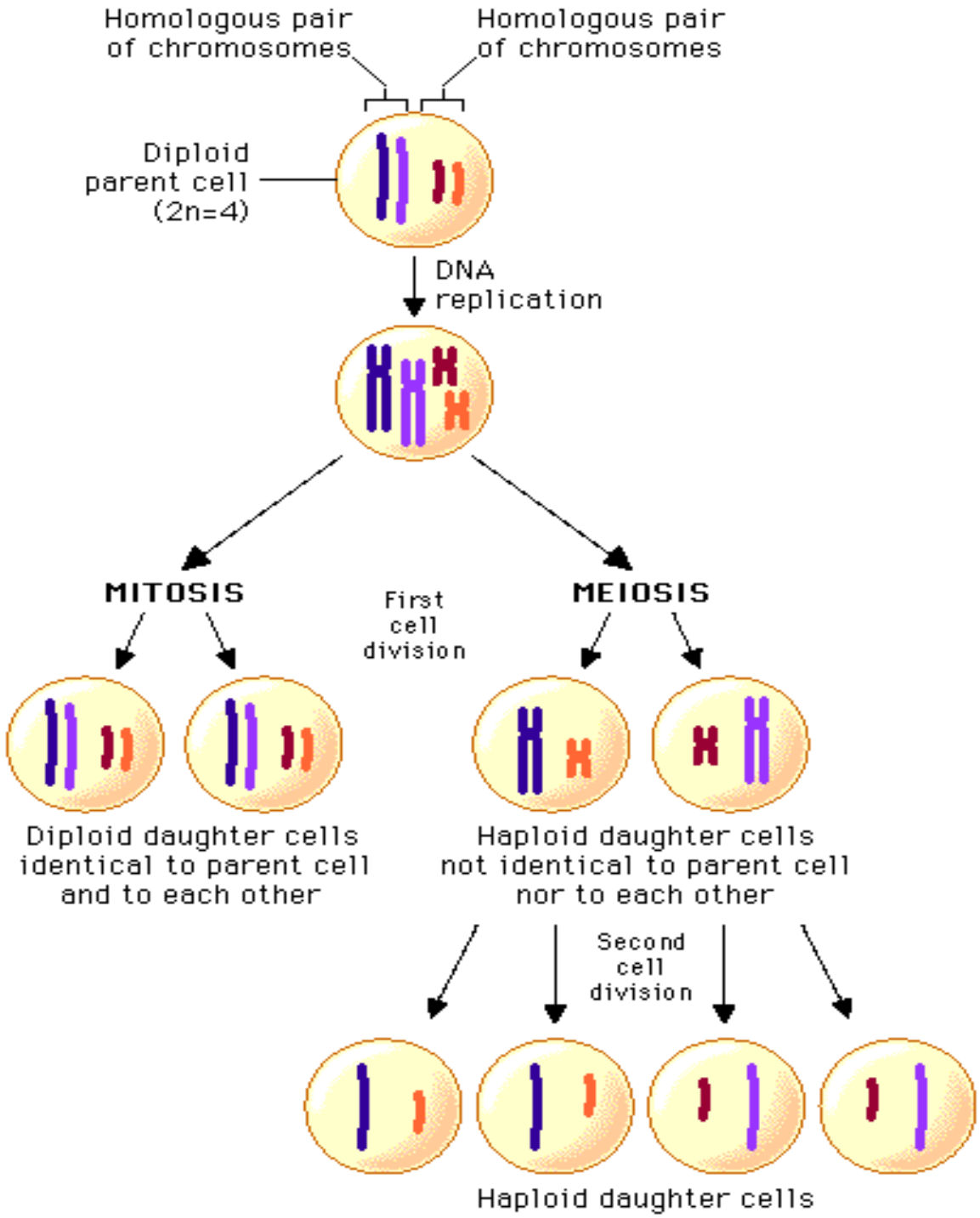
Questions

Select **all true** statements about meiosis.?

1. Meiosis involves two sequential cell divisions.
2. No DNA replication occurs before meiosis because the chromosome set is halved.
3. Crossing-over leads to variation in the genetic material.
4. During Meiosis I, paternal and maternal chromsomes are separated, ending up int two different cells.

The steps of Meiosis I and Meiosis II:




Homologous pair of chromosomes
Homologous pair of chromosomes
Diploid parent cell (2n=4)
DNA replication
MITOSIS
First cell division
MEIOSIS
Diploid daughter cells identical to parent cell and to each other
Haploid daughter cells not identical to parent cell nor to each other
Second cell division
Haploid daughter cells